

Animal kingdom

Basis of Classification

Multicellular
Eukaryotic
Holozoic mode of Nutrition

⇒ Metazoa

Parazoa

eg Pontera

Eumetazoa

eg - Coelenterata - Chordata

1) - Level of Organisation

Cellular — Pontera
Tissue — Coelenterata & Ctenophora
Organ — Platyhelminthus
Organ System — Aschelminthus to Chordata

2) - Pattern of Complexities

Digestive System

Incomplete



→ Coelenterata
Ctenophora
Platy.

Complete



→ Asch. → Chordata

open

closed

→ Arthropoda & Mollusca

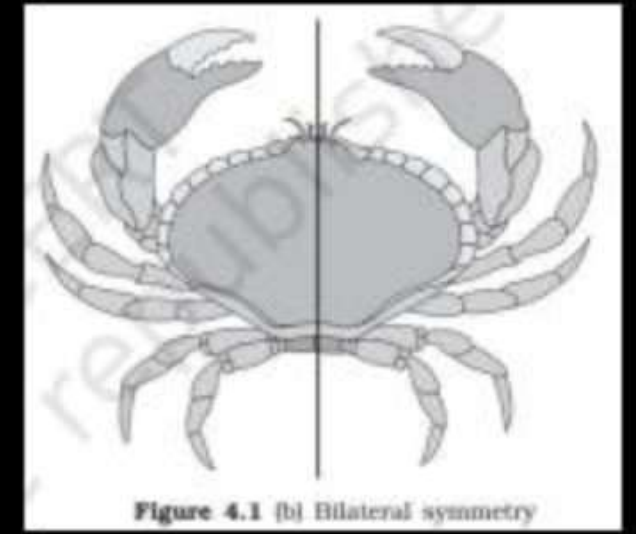
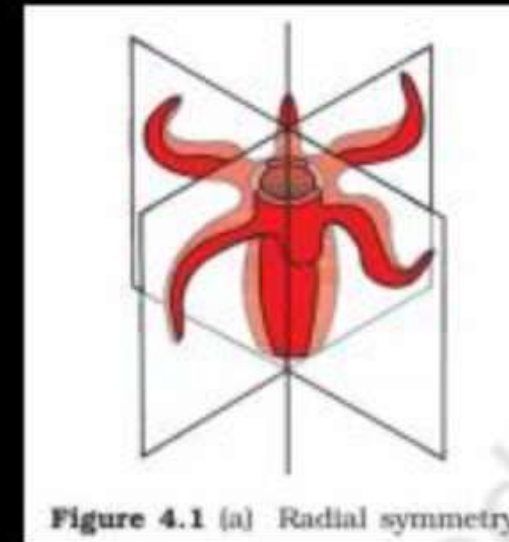
→ Annelida & Chordata

(3) Symmetry -

Radial → Some porifera, Coelenterata
Ctenophora, Adult Echino.

Bilateral → Platyhelminthes to Chordata

Asymmetry → Mostly Sponges eg- Spongilla

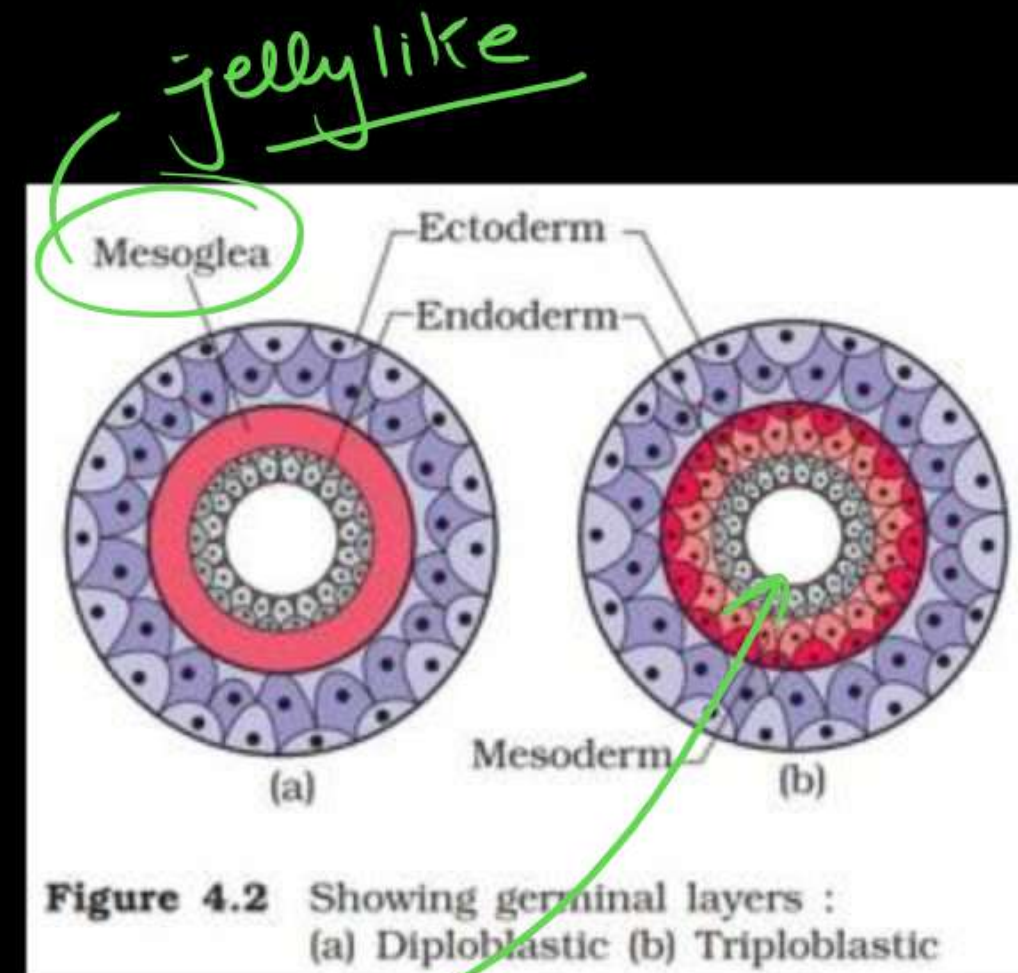


(4) Diploblastic and Triploblastic Organisation (Germ Layers)

Coelenterata
Ctenophora

Meso-
nt

Platyhelminthes
to
Chordata.



(5) Coelom - [Origin = Mesodermal → +nt b/w —
Outer body wall & Inner Gut Wall]

vs

Acoelomate

eg → Coelenterata
Ctenophora
Platyhelminths
Solid Mesoderm

Pseudocoelomate

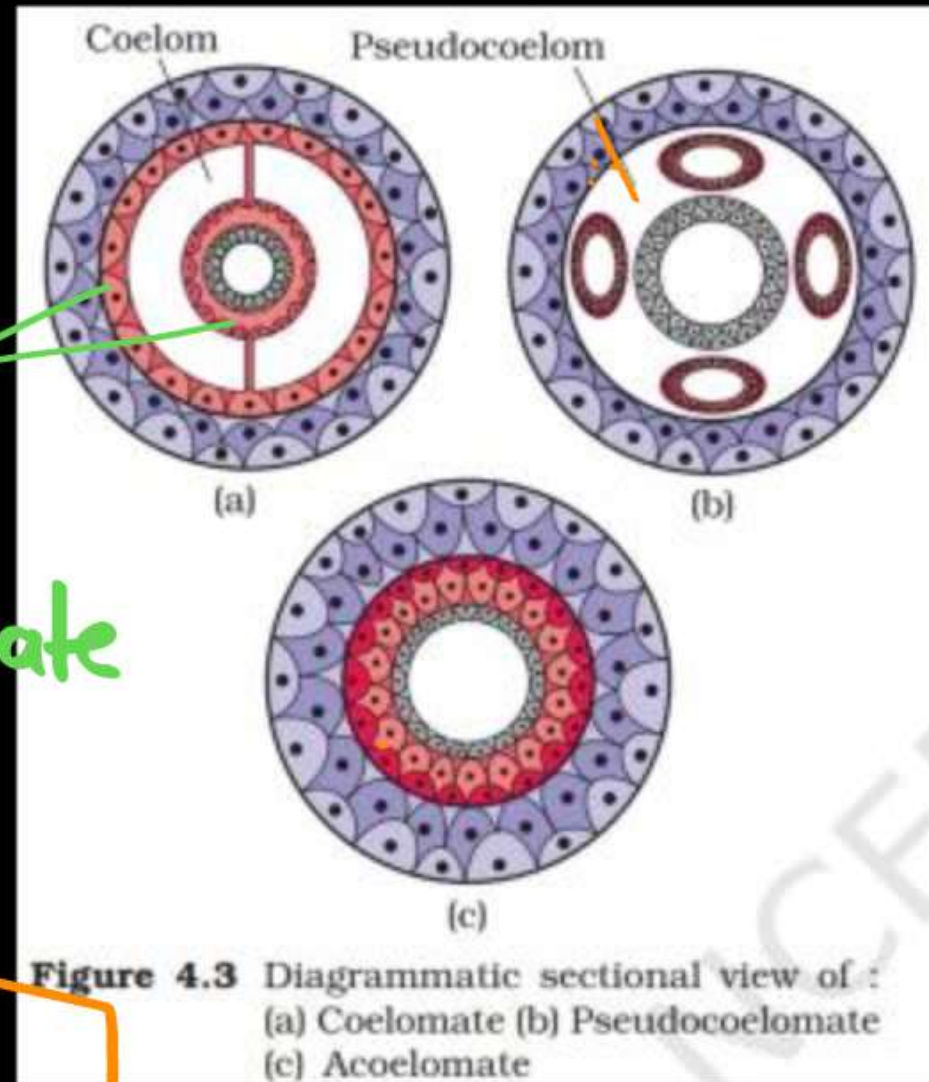
eg - Aschelminthes

Eucoelomate

Schizocoelom
[Splitting of Mesoderm]

eg → Annelida
Arthropoda
Mollusca

Entero.
[Outpocketing of Meso.]
Echinodermata
to
Chordata.



(7) Segmentation -

↓
[Metameric Segmentation]

Position of Internal Organ according to External Segmentation

= eg → Annelida ✓
→ Arthropoda ✓
→ Chordata ✓

(8) Notochord -

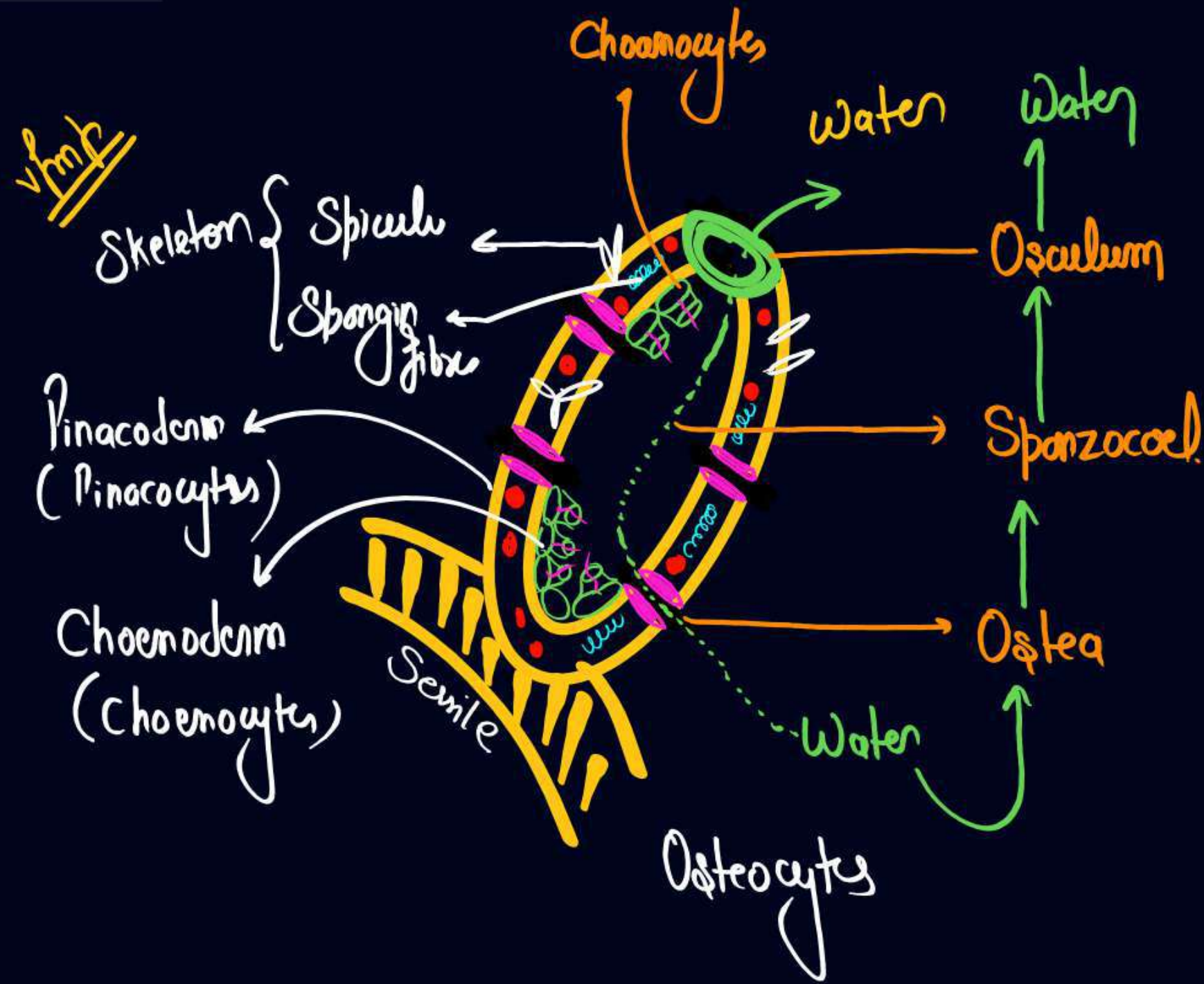
↓
Origin =
Mesodermal
(Endoskeleton)

→ Nonchordata = Notochord - nt

→ Chordata = Notochord + nt

In some Chordata Notochord Replace by
V. Column Called = Vertebrata

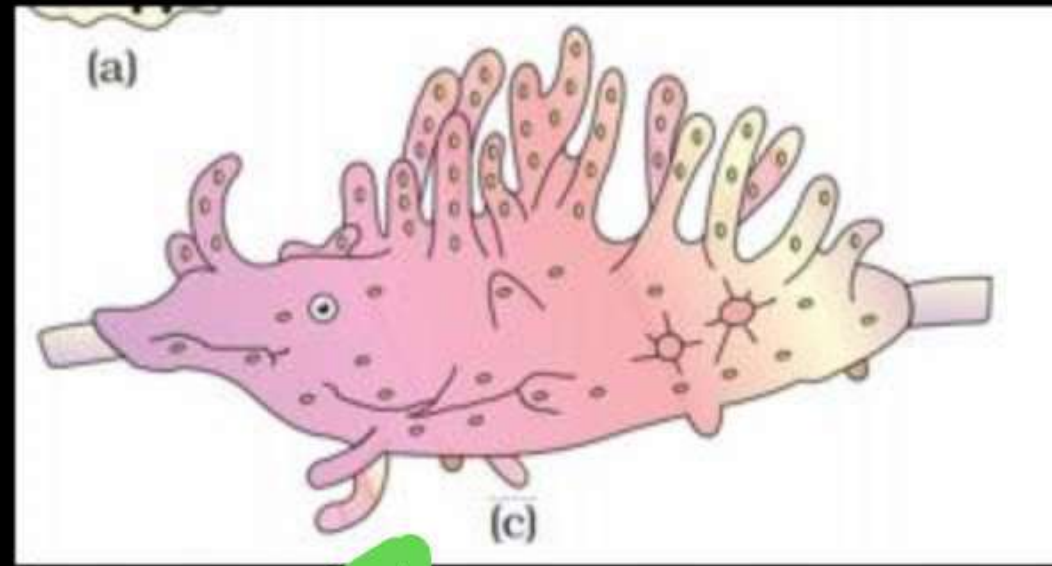
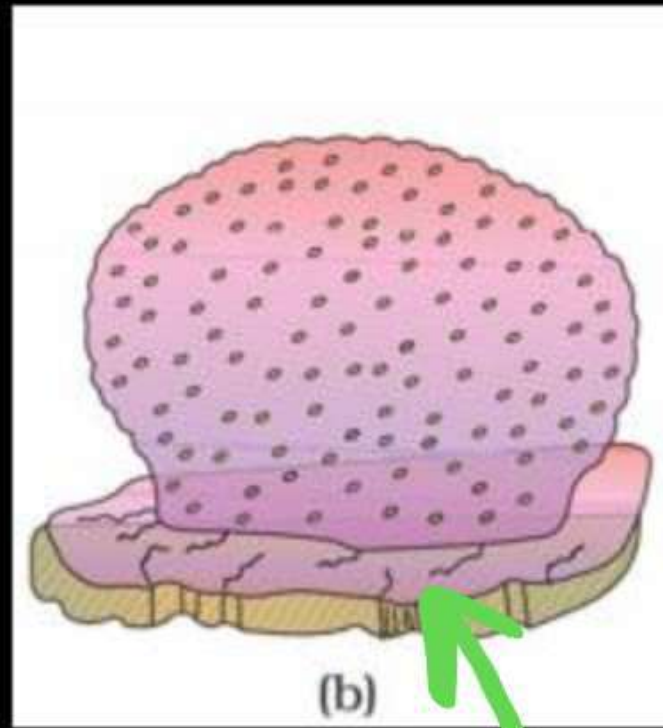
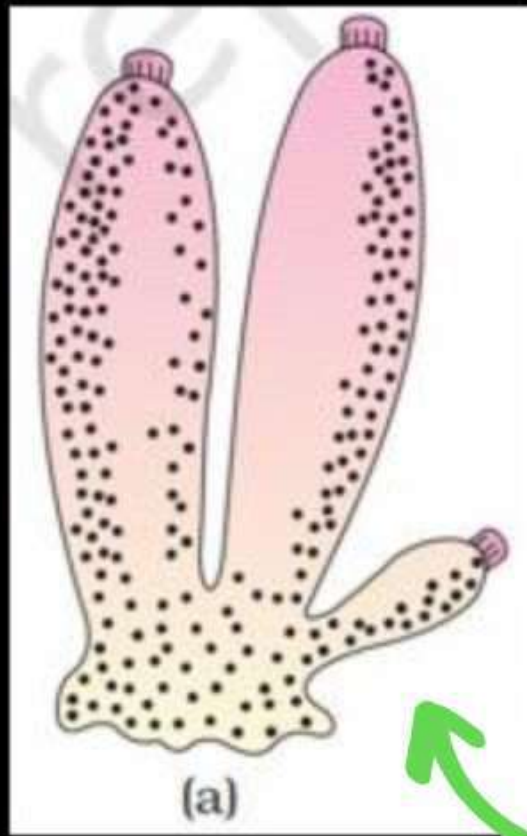
Porifera



~~Vamp~~ Water Canal System

Function

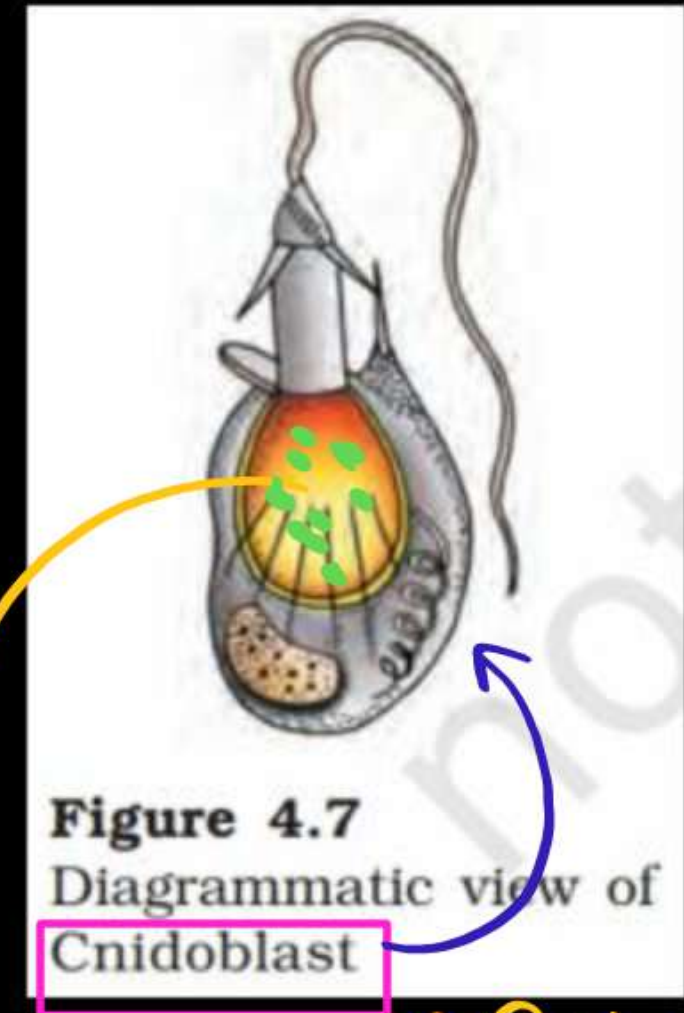
- Respiration ✓
- Excretion ✓
- Food Capture ✓
- Fertilization ✓
- etc.



Examples: Sycon (Scypha), Spongilla (Fresh water sponge) and Euspongia (Bath sponge).

Phylum – Coelenterata (Cnidaria)

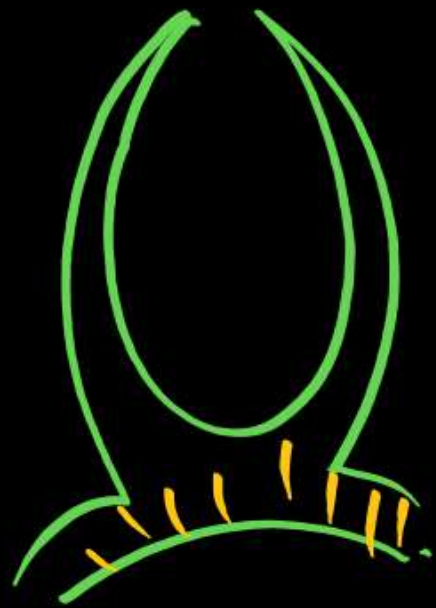
- . The name cnidaria is derived from the **cnidoblasts or cnidocytes** (which contain the stinging capsules or **nematocysts**) present on the tentacles and the body.
- . Cnidoblasts are used for **anchorage, defense** and for the **capture of prey**.
- . Some of the cnidarians, e.g., corals have a skeleton composed of calcium carbonate.



Nimatouyst

or Cnidocyst

vvv Members of this phylum show Bimorphic Condition



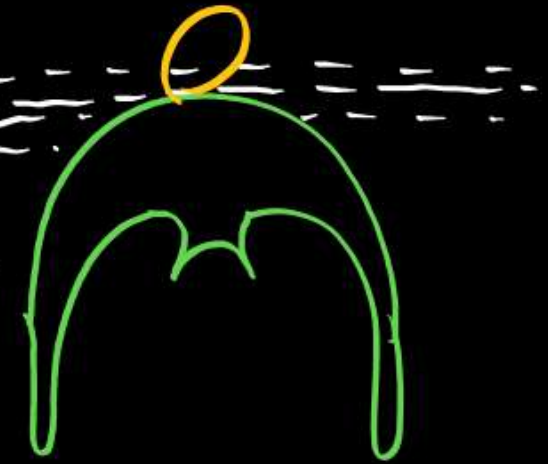
[Polyp]

- * Cylindrical
- * Sessile
- * Asexual Reproduction

eg = Hydra ✓
Obelia ✓
Adamsia ✓

[Medusa]

- * Umbrella like →
- * Free Swimming
- * Sexual Reproduction



eg = Aurelia
[jelly fish]



Examples: Physalia (Portuguese man-of-war), Adamsia (Sea anemone), Pennatula (Sea-pen), Gorgonia (Sea-fan) and Meandrina (Brain coral).



Phylum – Ctenophora

- Sea Walnut / Comb jelly — 8 Comb plates (Ciliated)
↓
Help in Locomotion

- Reproduction only Sexual

- Bioluminescence +nt

+nt only in Marine Water

Examples:

Ctenoplana.

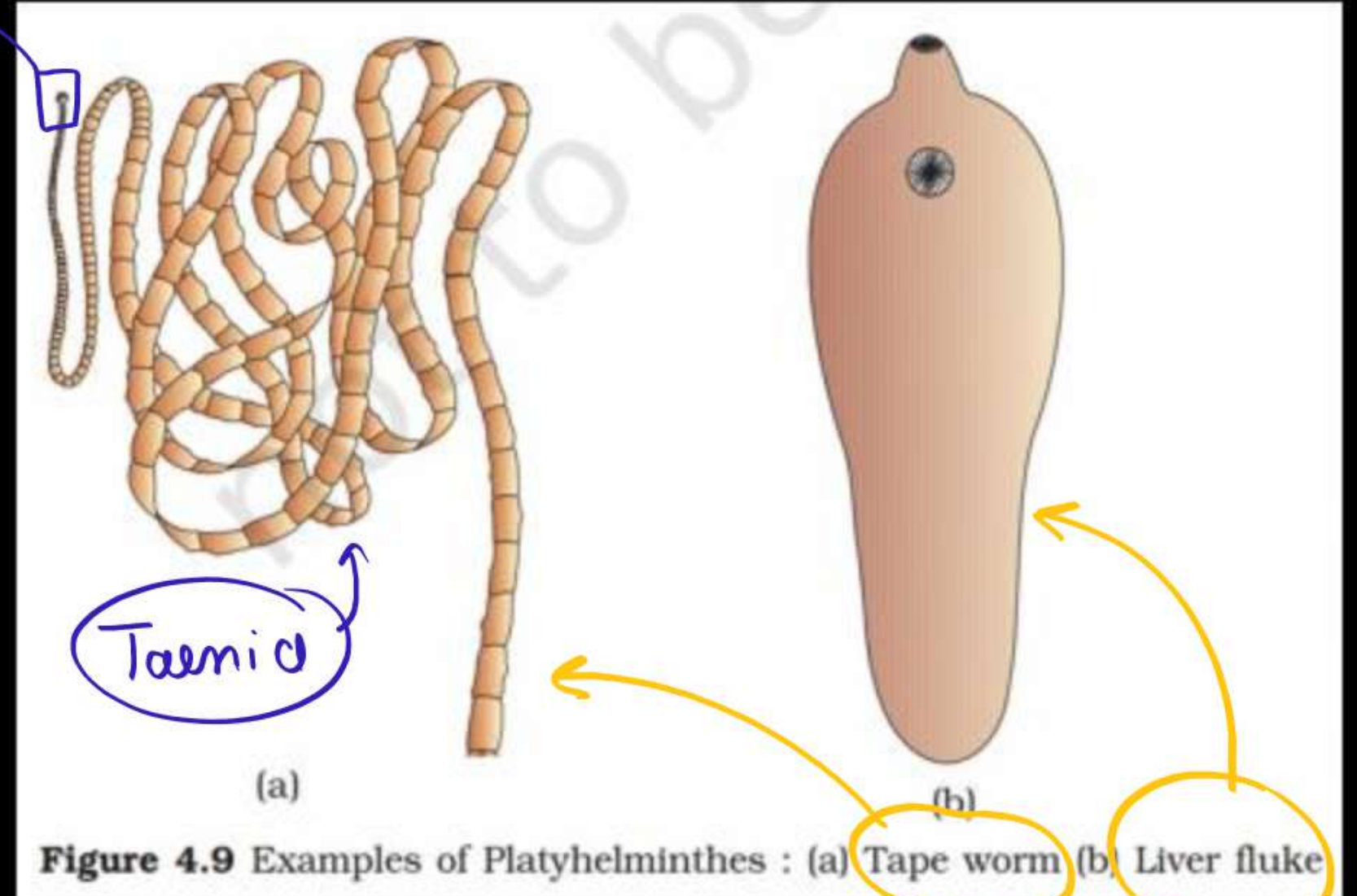
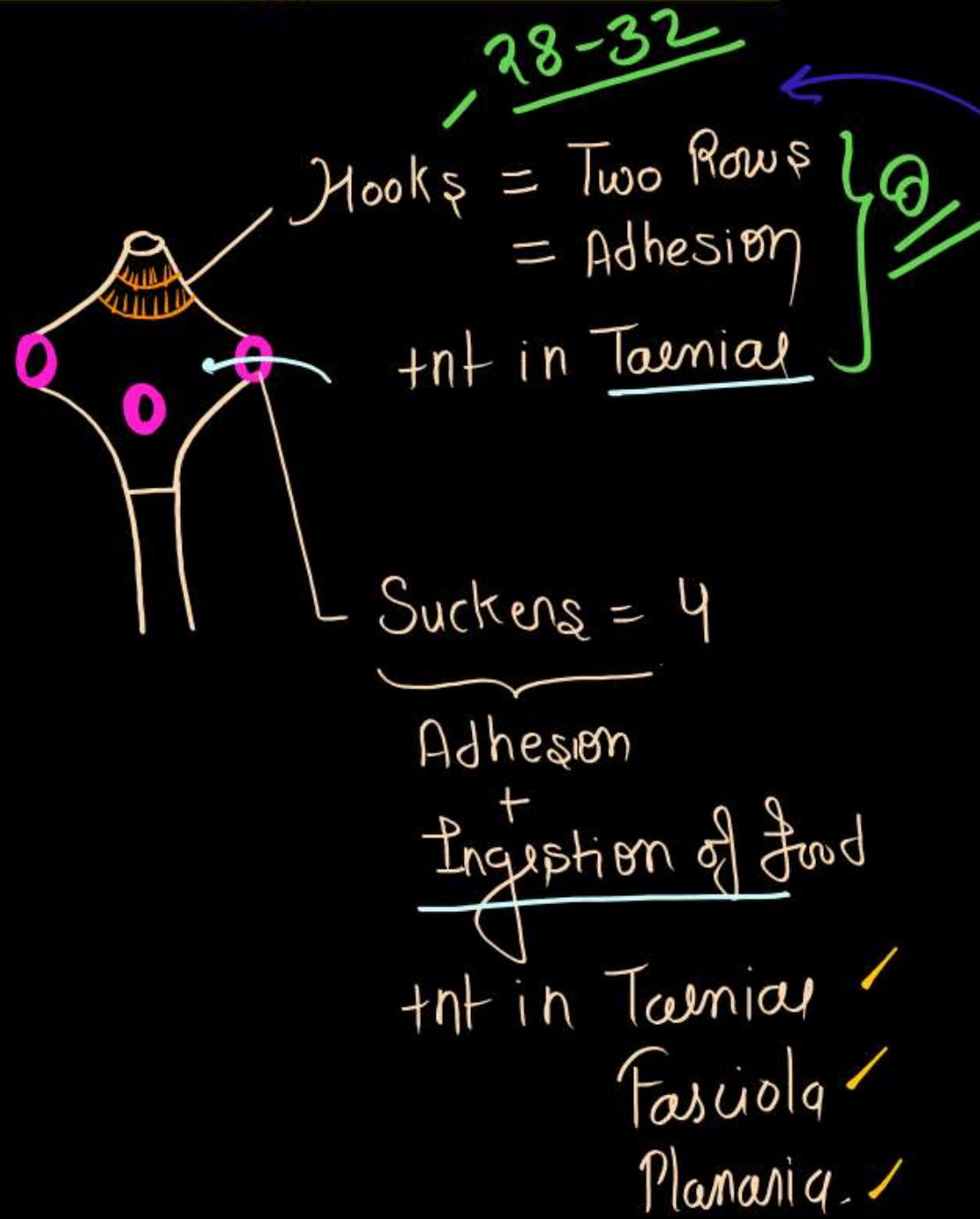
Pleurobrachia

and



Figure 4.8 Example of Ctenophora (Pleurobrachia)

Phylum – Platyhelminthes



eg - Planaria = High Regeneration

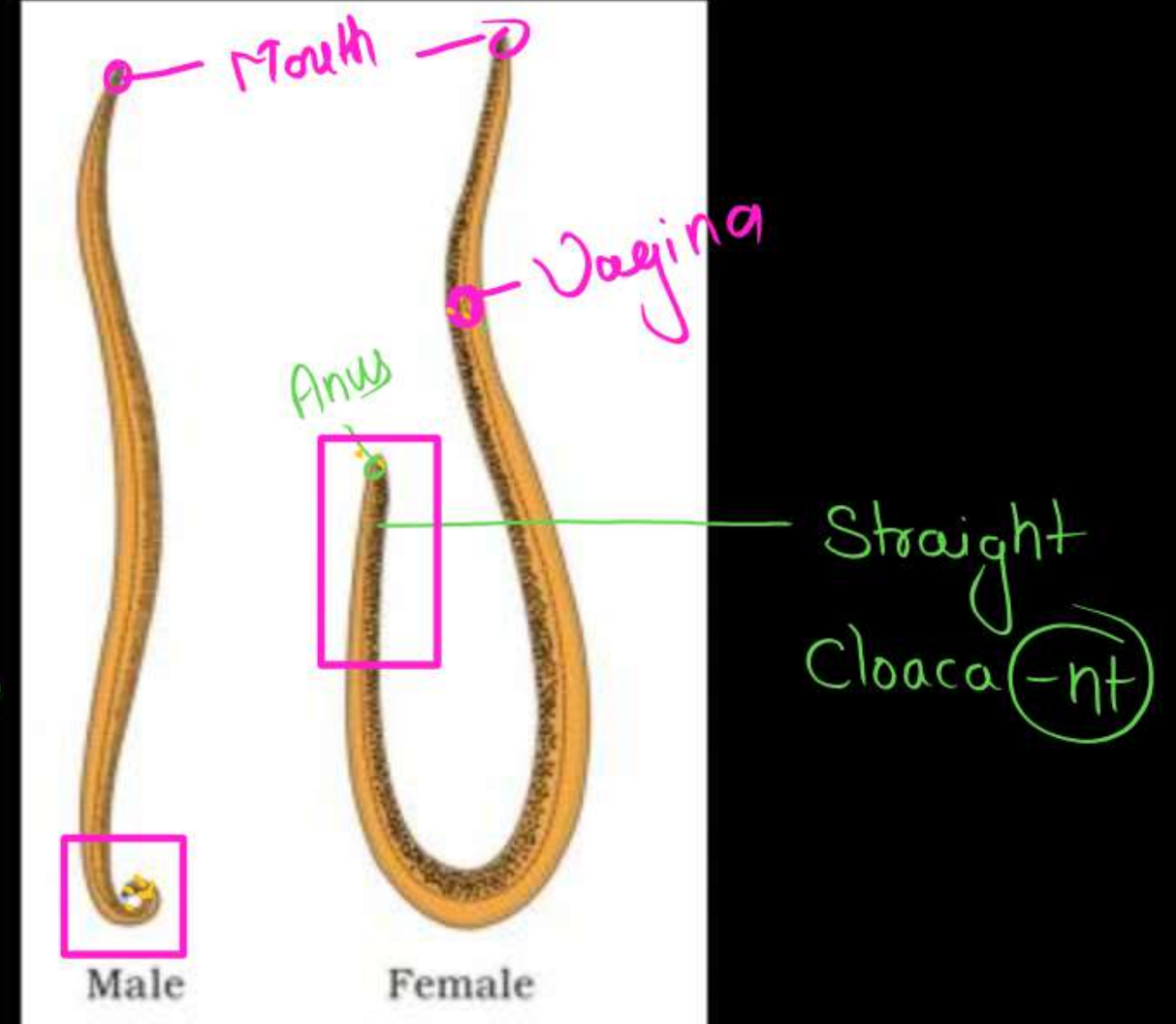
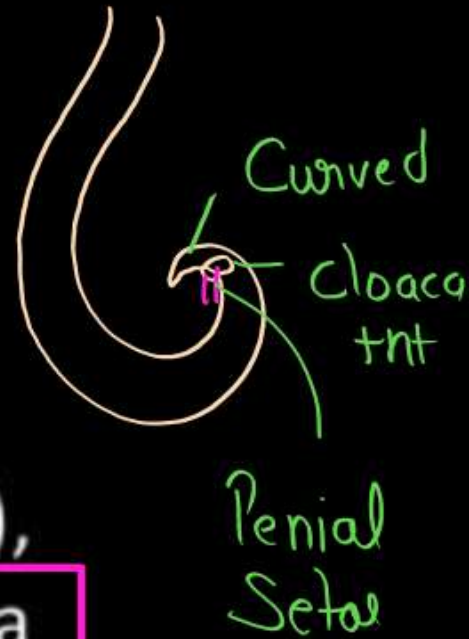
Phylum – Aschelminthes

Sexes are separate (dioecious), i.e., males and females are distinct. Often females are longer than males.

Examples

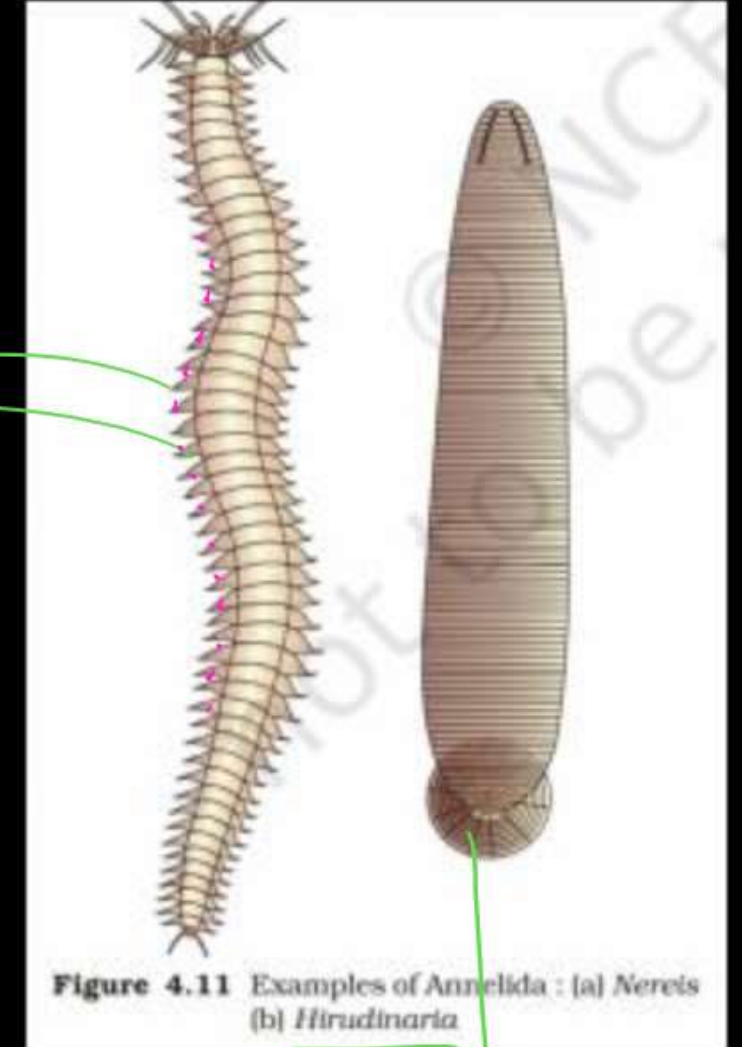
Ascaris (Roundworm),

Wuchereria (Filaria worm), Ancylostoma (Hookworm).



Phylum – Annelida

② Parapodia
├── Locomotion
└── Respira.



✓
Examples : *Nereis*, *Pheretima* (Earthworm) and *Hirudinaria* (Blood sucking leech).

Phylum – Arthropoda (arthros-joint, poda-appendages)

1. This is the largest phylum of Animalia which includes insects.
2. Over two-thirds of all named species on earth are arthropods.

70%

Examples: Economically important insects –

Apis (Honey bee),

Bombyx (Silkworm),

Laccifer (Lac insect)

Vectors – Anopheles, Culex and Aedes
(Mosquitoes)

Gregarious pest – Locusta (Locust)

Limulus – living fossil

vvvvv fomp

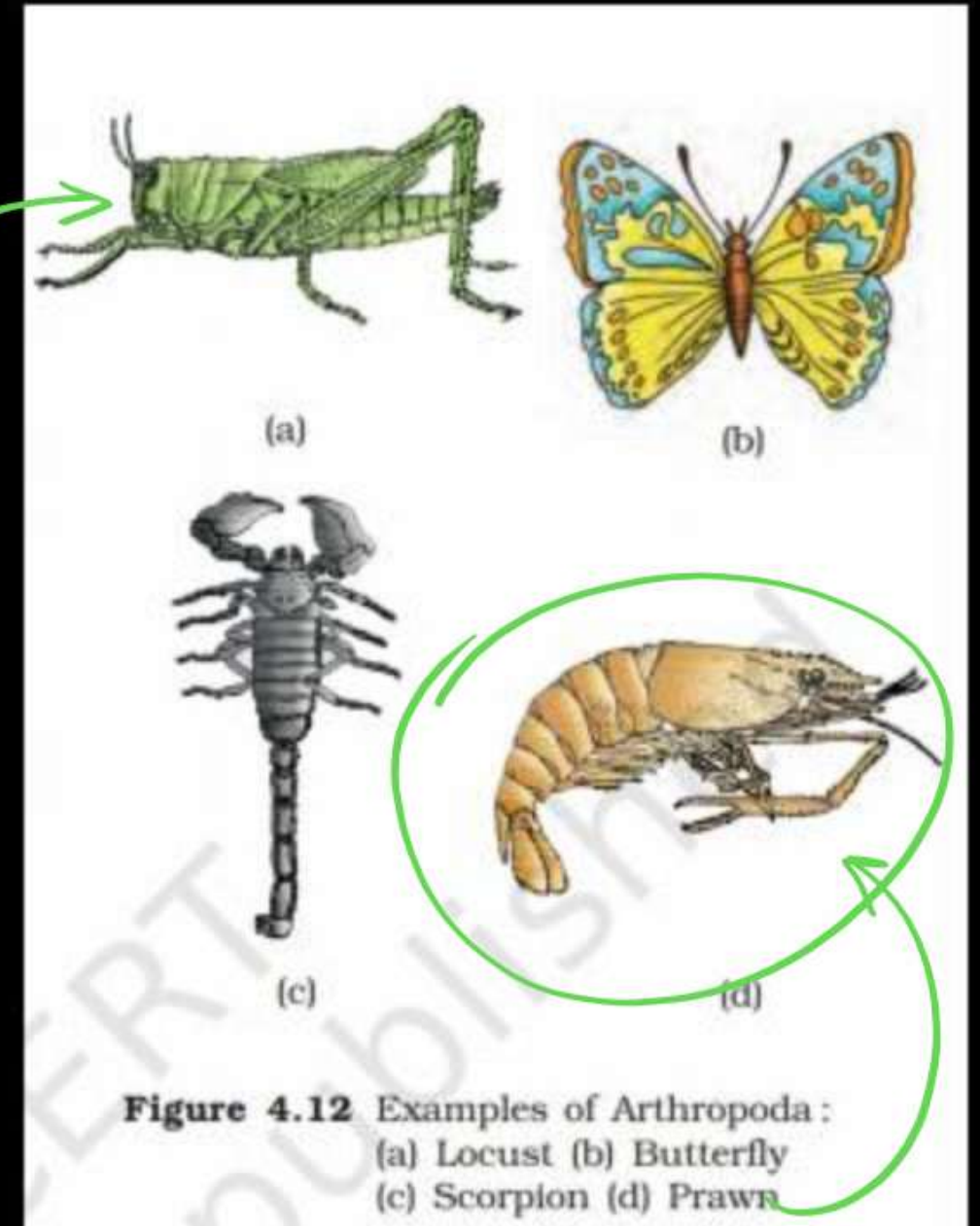
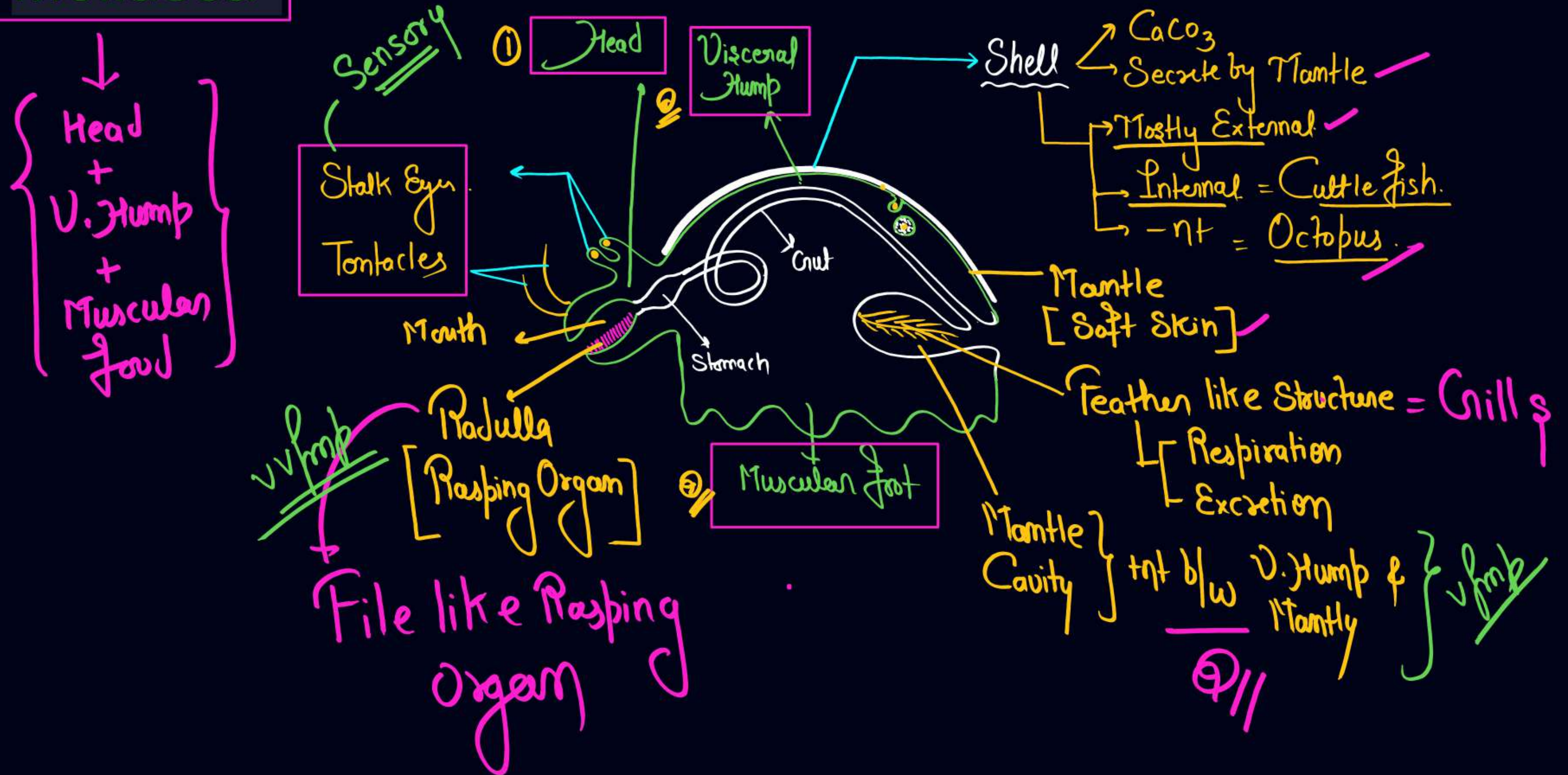
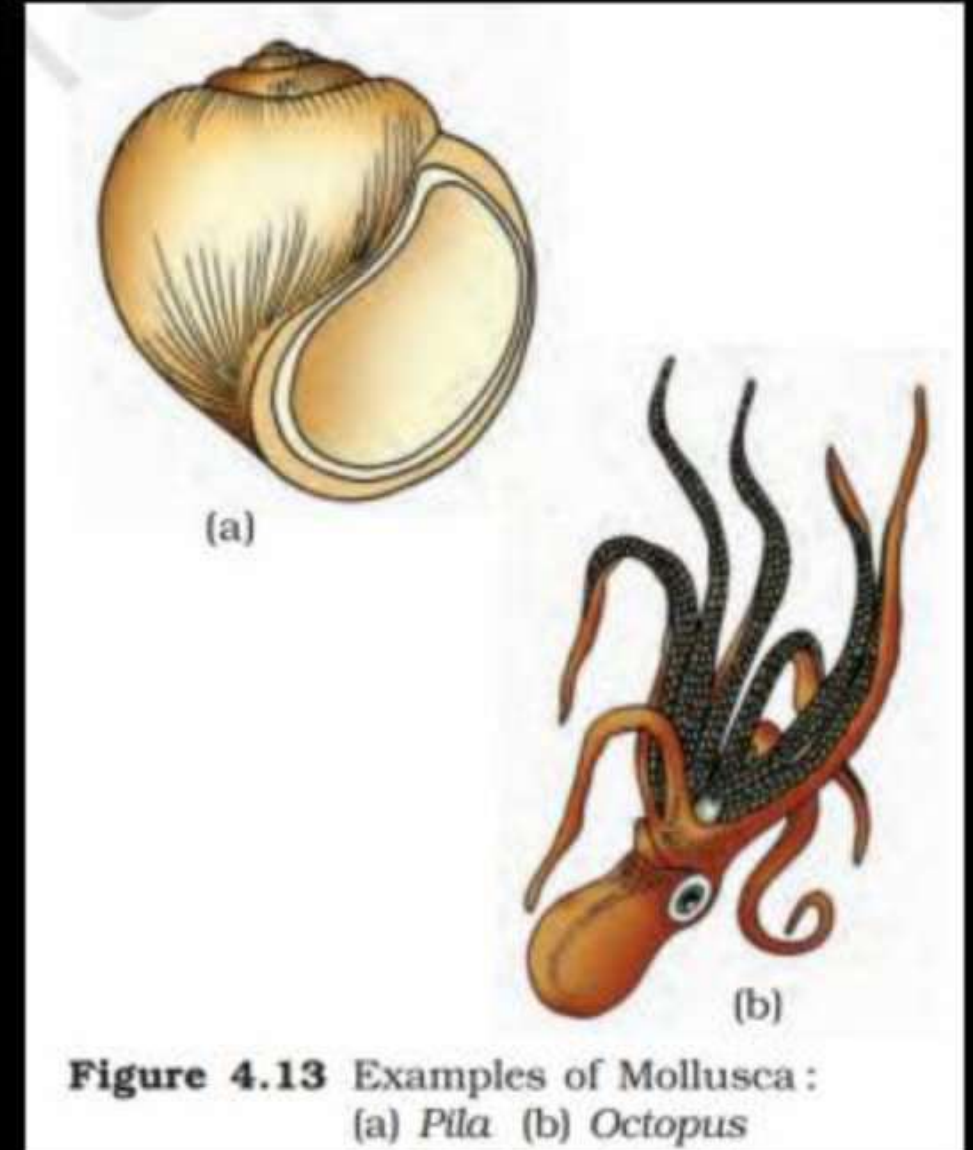


Figure 4.12 Examples of Arthropoda :
(a) Locust (b) Butterfly
(c) Scorpion (d) Prawn

Mollusca

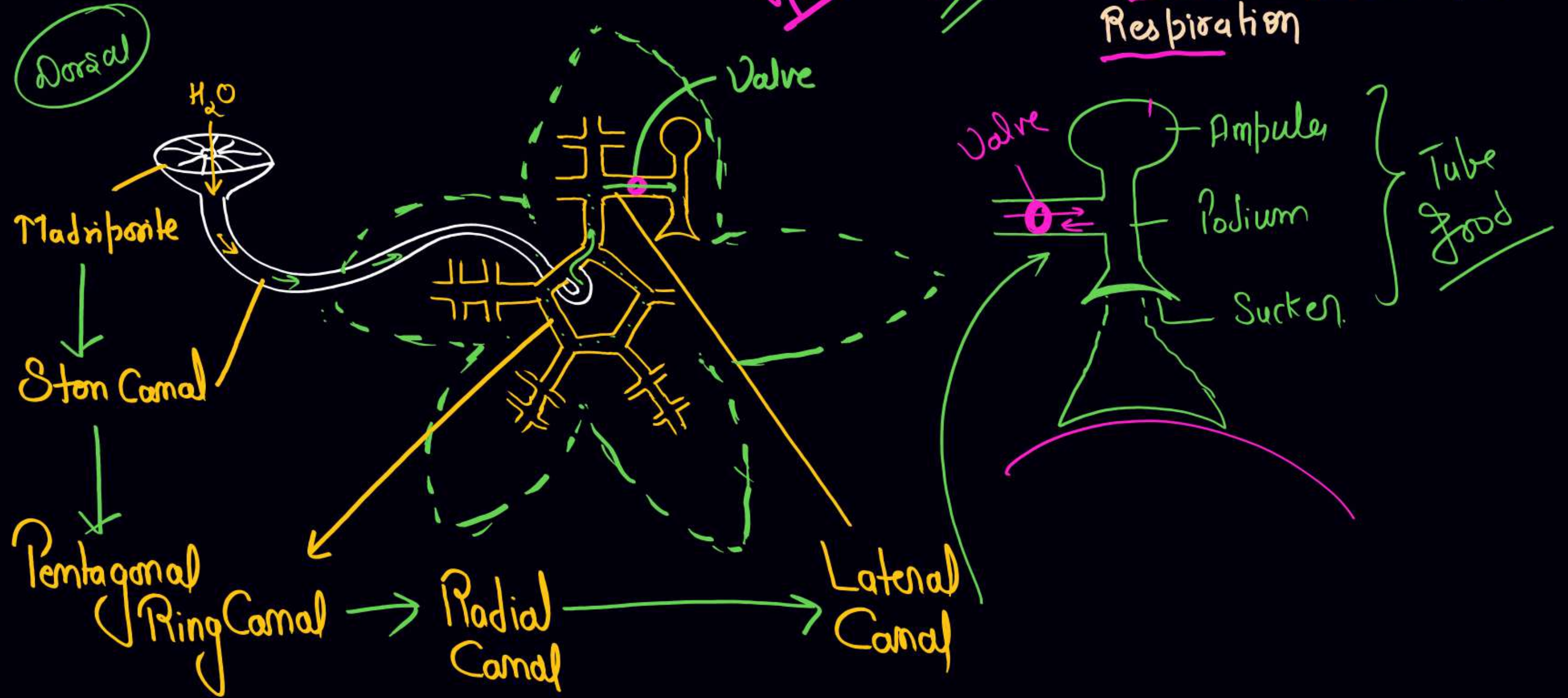


Examples: Pila (Apple snail), Pinctada (Pearl oyster), Sepia (Cuttlefish), Loligo (Squid), Octopus (Devil fish), Aplysia (Sea hare), Dentalium (Tusk shell) and Chaetopleura (Chiton).



Echinodermata

= Water Vascular System = Locomotion /
Capture & Transport of food
Respiration



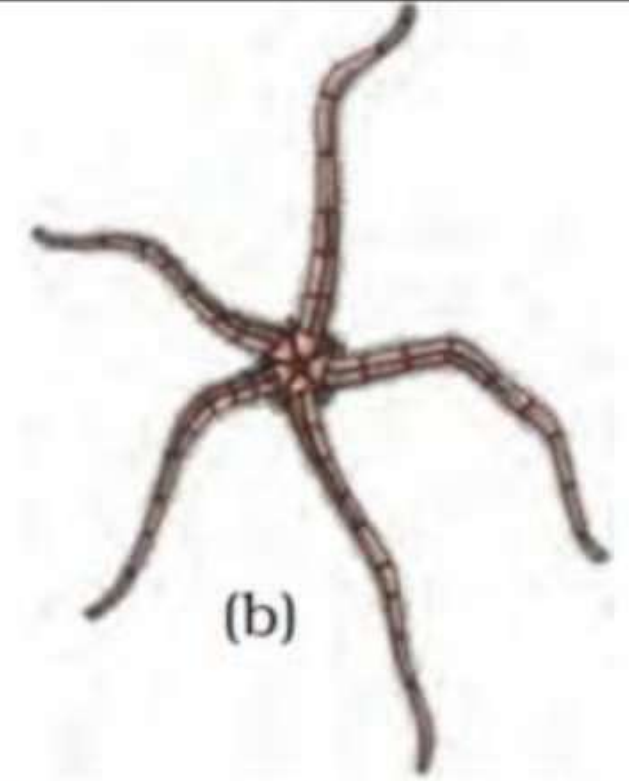
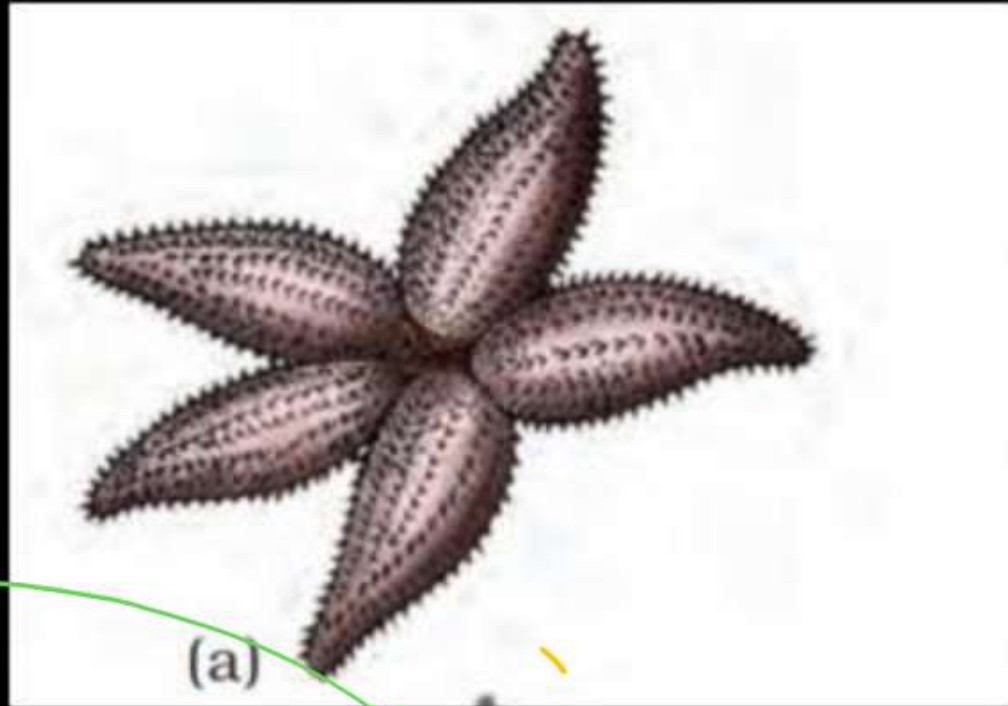
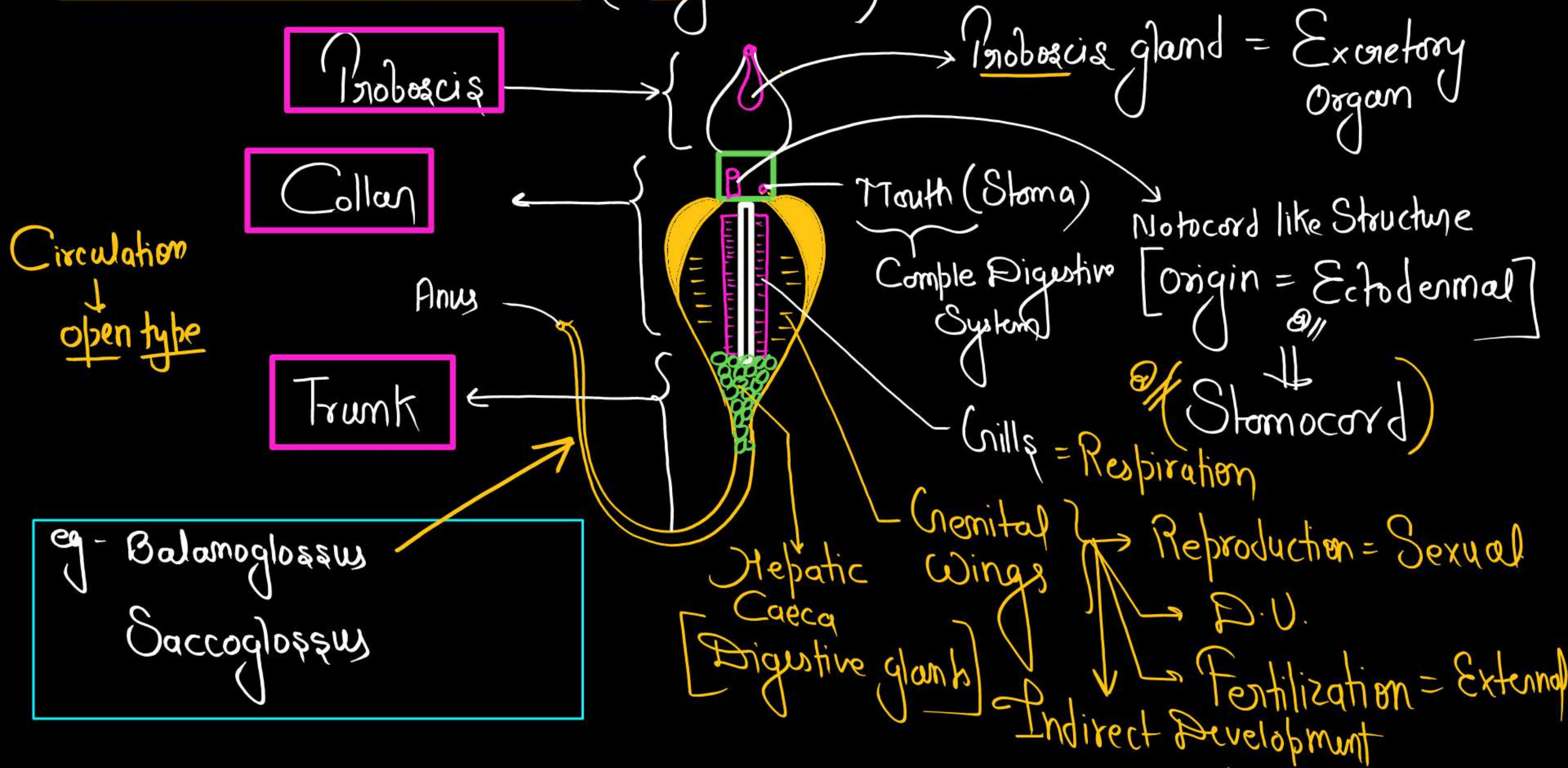


Figure 4.14 Examples of Echinodermata :
(a) *Asterias*
(b) *Ophiura*

Examples: Asterias (Star fish), Echinus (Sea urchin), Antedon (Sea lily), Cucumaria (Sea cucumber) and Ophiura (Brittle star).

Phylum – Hemichordate (Tongue Worm)



Chordata

Origin - Ectodermal
+ nt

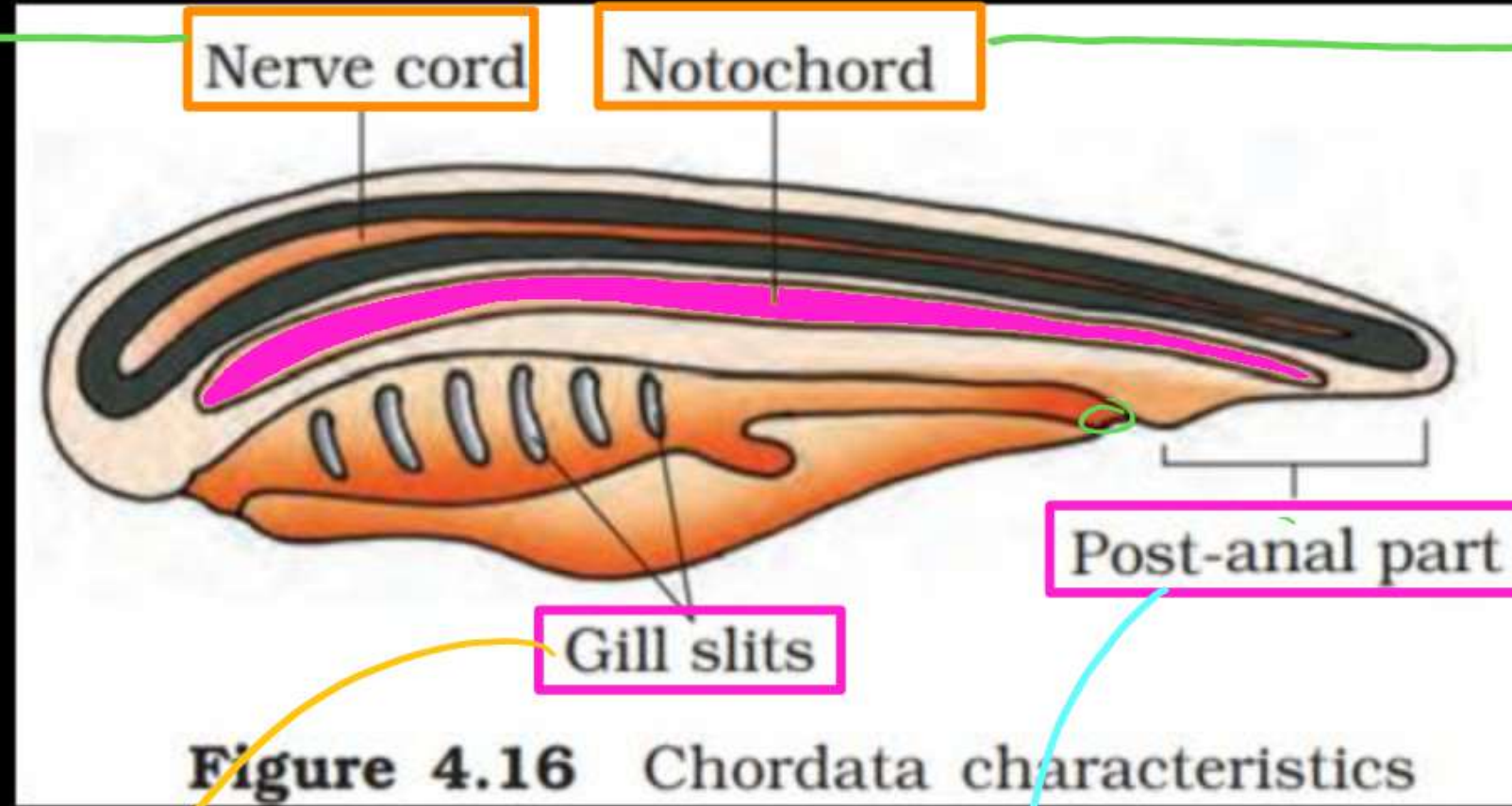
Nonchordates

Chordates

Ventral
Double
Solid

Dorsal
Single
Hollow

In some higher
Chordate Gill Slits
+ nt only in Embryonic
Stage.



Origin = Mesodermal
+ nt only in chordata
Supporting Structure
+ nt behind = Nerve Cord
above = Gut

for Balancing

In-Some chordate.
Notochord Replace by
V. Columns

www.fmp2

Phylum Chordata

(Brain box)

www.fmp2

Protochordata [Acraniata]

(Sub-Phylum)

Urochordata

Tail

* Notochord + nt only
in tail of Larva —
-nt in Adult

eg → Ascidia
Solpa
Doleolum

Cephalochordata

Head

* Notochord + nt —
Head to Tail in —
Larva & Adult both

eg → Branchiostoma
or
Amphioxus or lanulelet

Euchordata [Craniata]

Sub-Phylum

Vertebrata

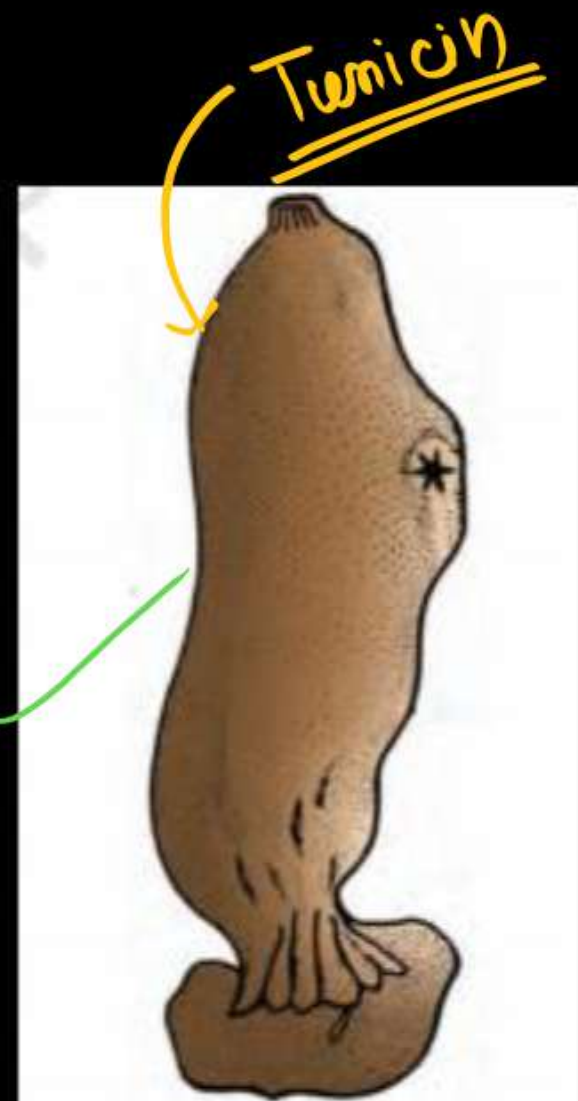
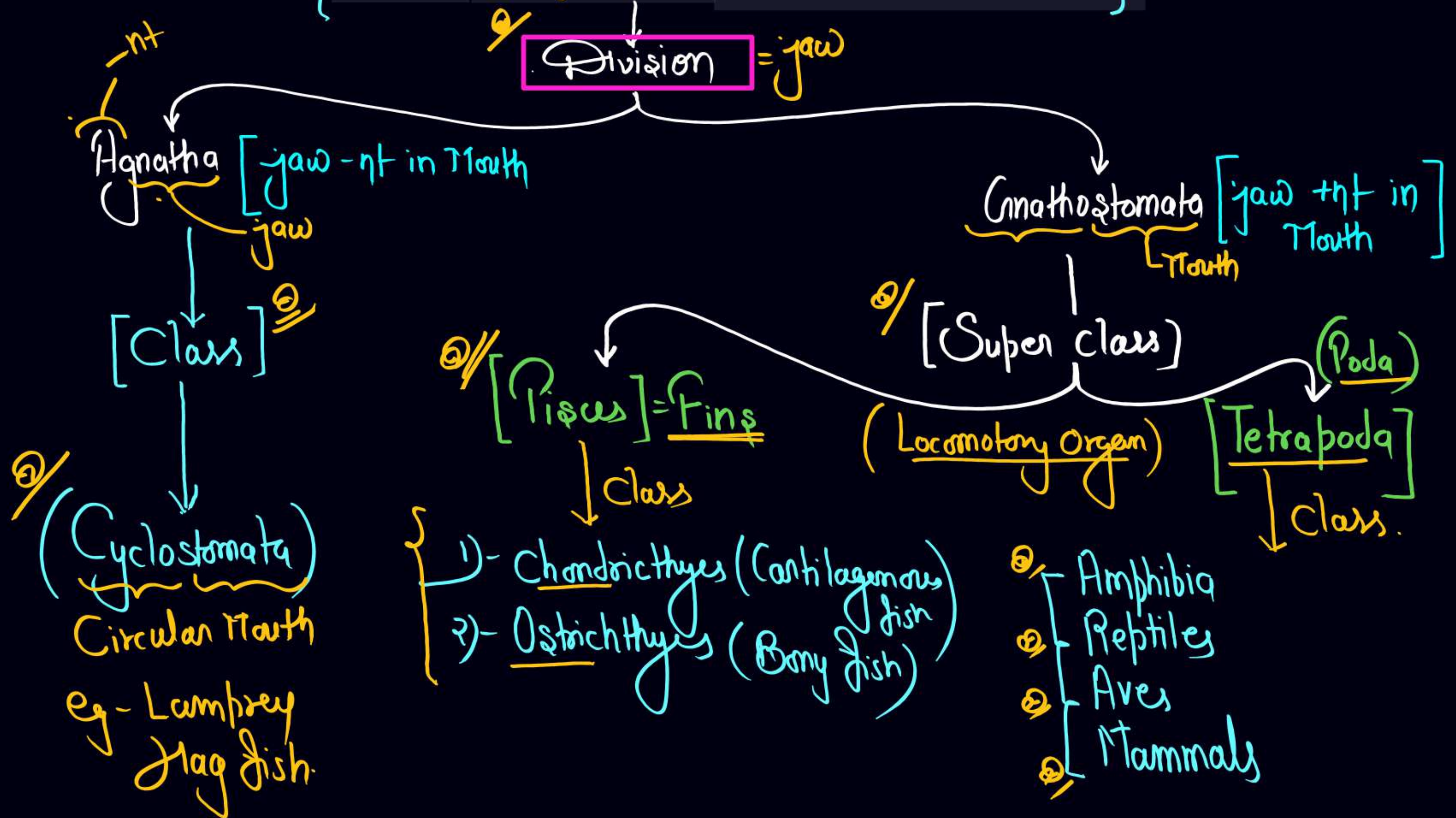


Figure 4.17 Ascidia

{ Sub-Phylum = Vertebrata }



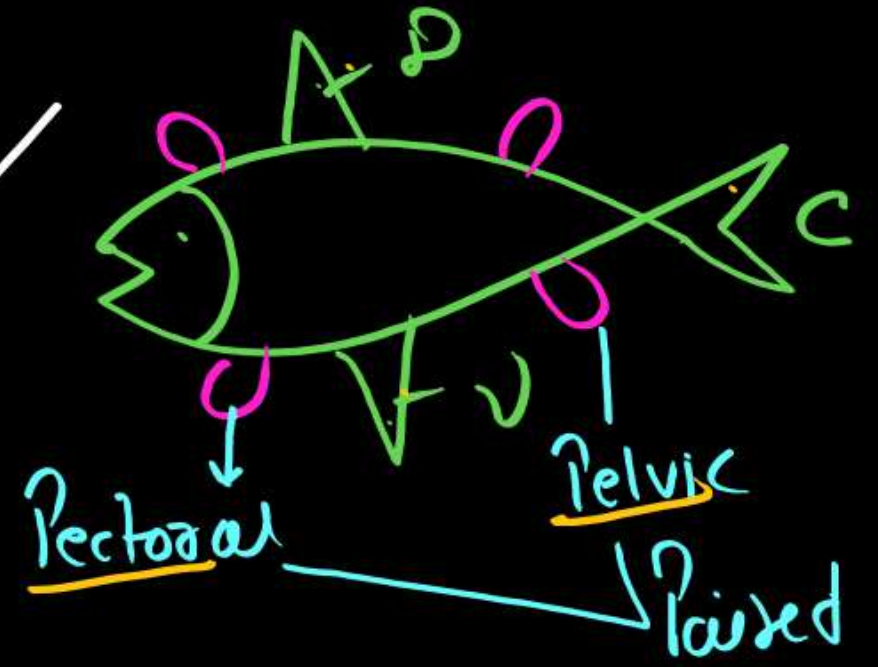
① Class – Cyclostomata

Circular Mouth

vvvmp

1. All living members of the class Cyclostomata are ectoparasites on some fishes.
2. They have an elongated body bearing 6-15 pairs of gill slits for respiration.
3. Cyclostomes have a sucking and circular mouth without jaws.
4. Their body is devoid of scales and paired fins. unpaired fins +nt
5. Cranium and vertebral column are cartilaginous. ✓
6. Circulation is of closed type.

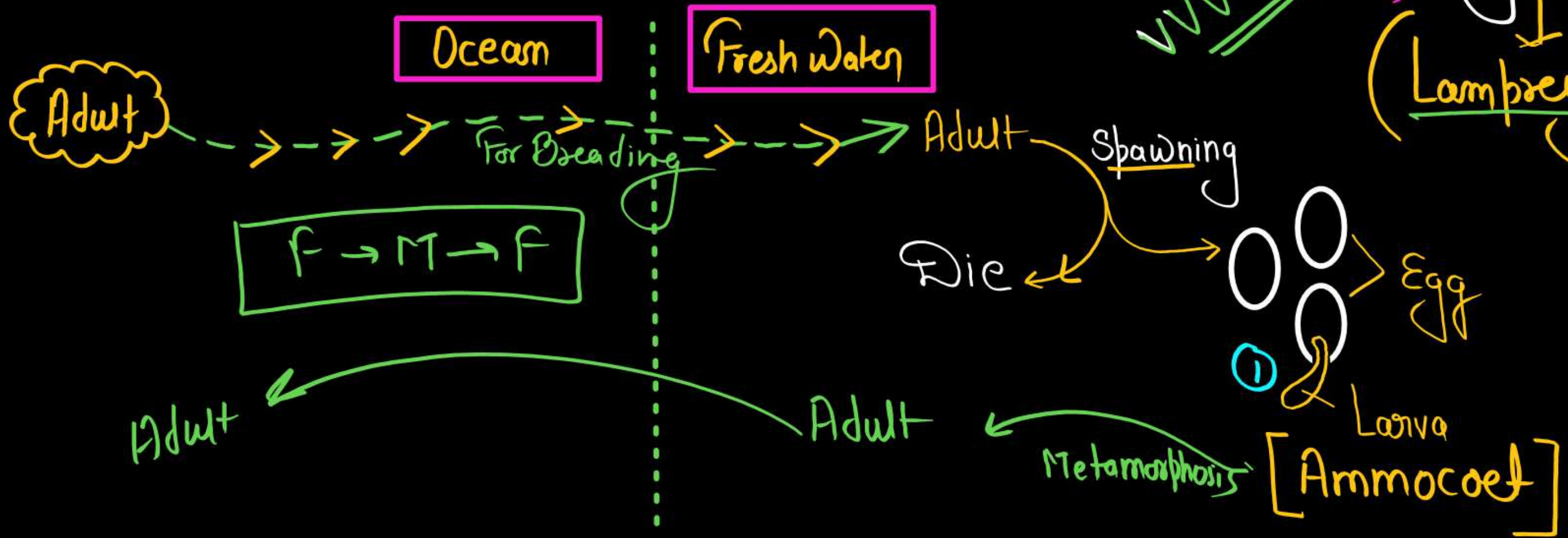
vvvmp



Cyclostomes are marine but migrate for spawning to fresh water.
After spawning, within a few days, they die.
Their larvae, after metamorphosis, return to the ocean.

Anadromous
Migration

(Lamprey + H)



Examples: Petromyzon (Lamprey) and Myxine (Hagfish).

Cartilaginous

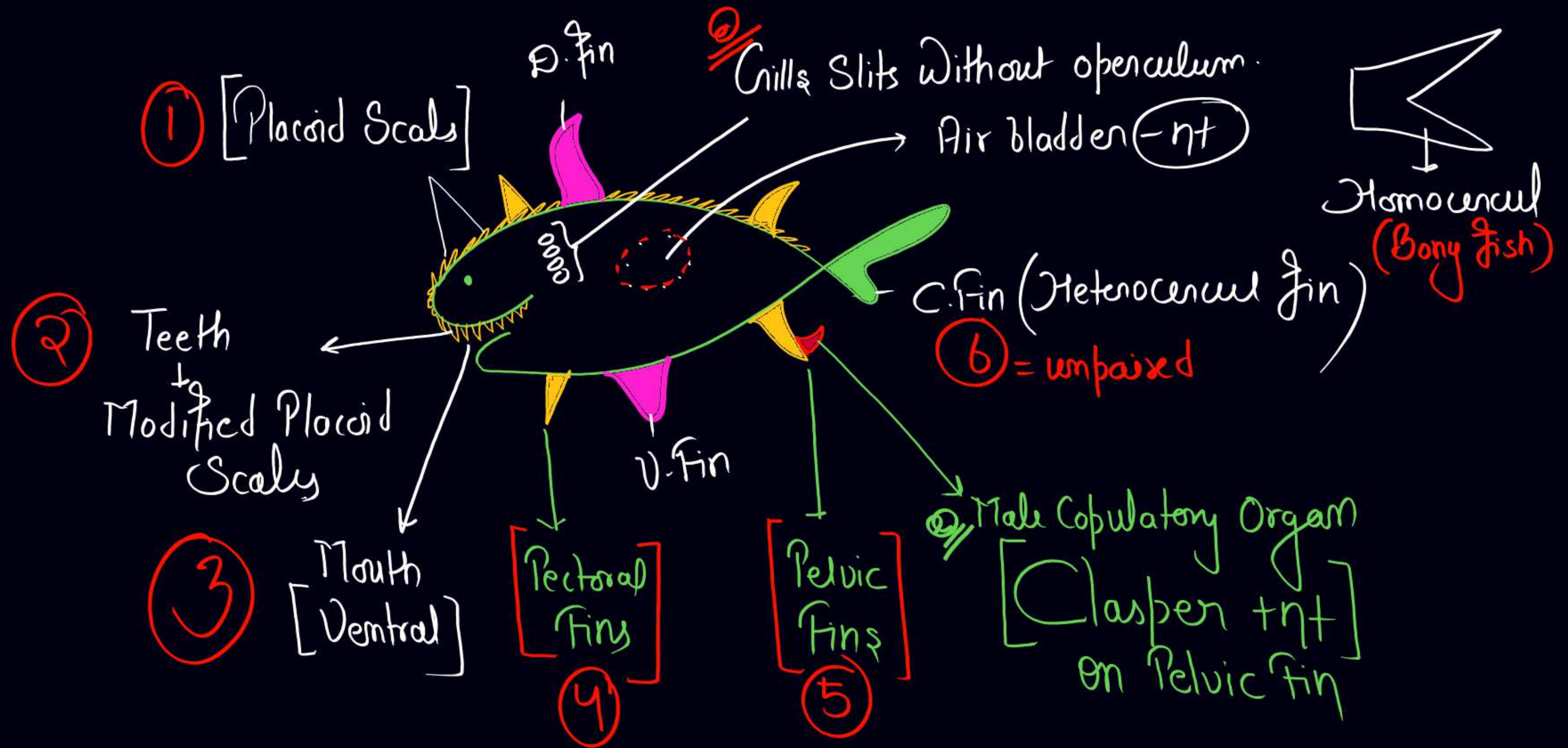
Class – Chondrichthyes

1. They are marine animals with streamlined body and have cartilaginous endoskeleton.
2. Mouth is located ventrally.
3. Notochord is persistent throughout life.
4. Gill slits are separate and without operculum (gill cover).
5. The skin is tough, containing minute placoid scales. Teeth are modified placoid scales which are backwardly directed.
6. Their jaws are very powerful.
7. These animals are predaceous.

Bony Fish.

Class – Osteichthyes

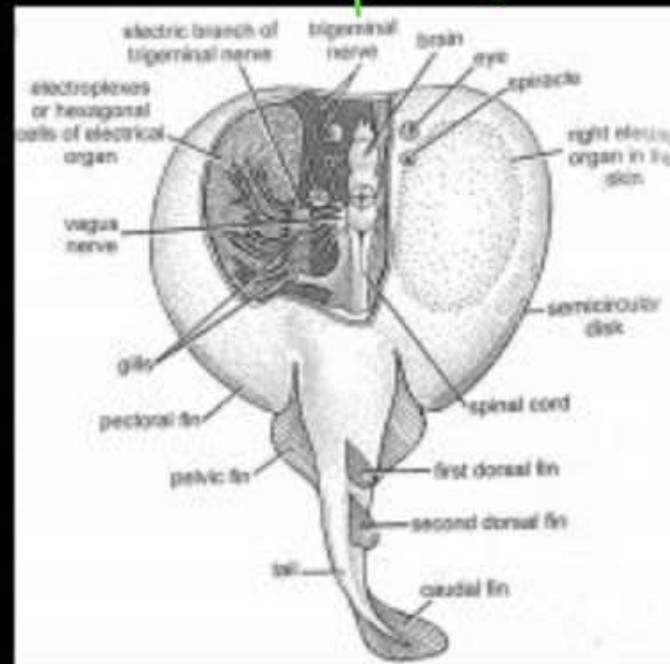
1. It includes both marine and fresh water fishes with bony endoskeleton. Their body is streamlined.
2. Mouth is mostly terminal.
3. They have four pairs of gills which are covered by an operculum on each side.
4. Skin is covered with cycloid/ctenoid scales.
5. Air bladder is present which regulates buoyancy.
6. Heart is two chambered (one auricle and one ventricle).
7. They are cold-blooded animals.



vvvvvmp

Examples: Scoliodon (Dog fish), Pristis (Saw fish), Carcharodon (Great white shark), Trygon (Sting ray).

Torpedo (Electric Ray)



Trigem (Sting ray)

Examples:

● Marine – Exocoetus (Flying fish), Hippocampus (Sea horse);

Freshwater – Labeo (Rohu), Catla (Katla), Clarias (Magur); Aquarium – Betta (Fighting fish), Pterophyllum (Angel fish)

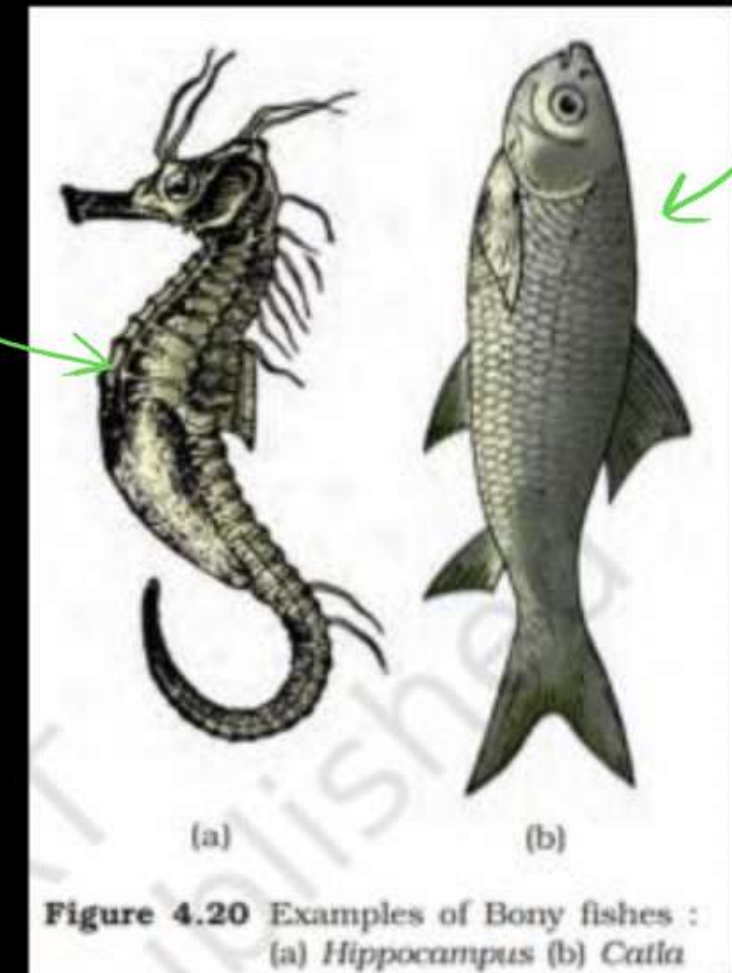
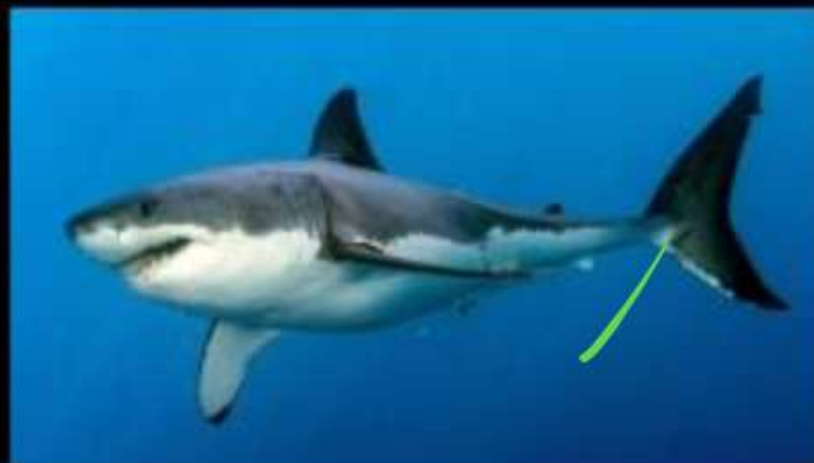
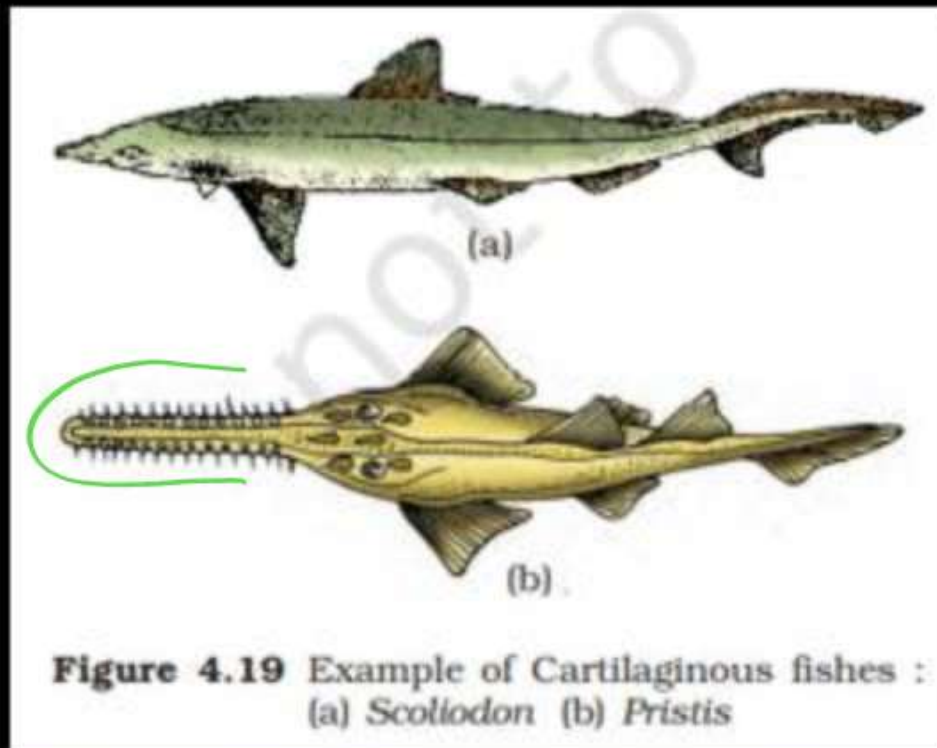
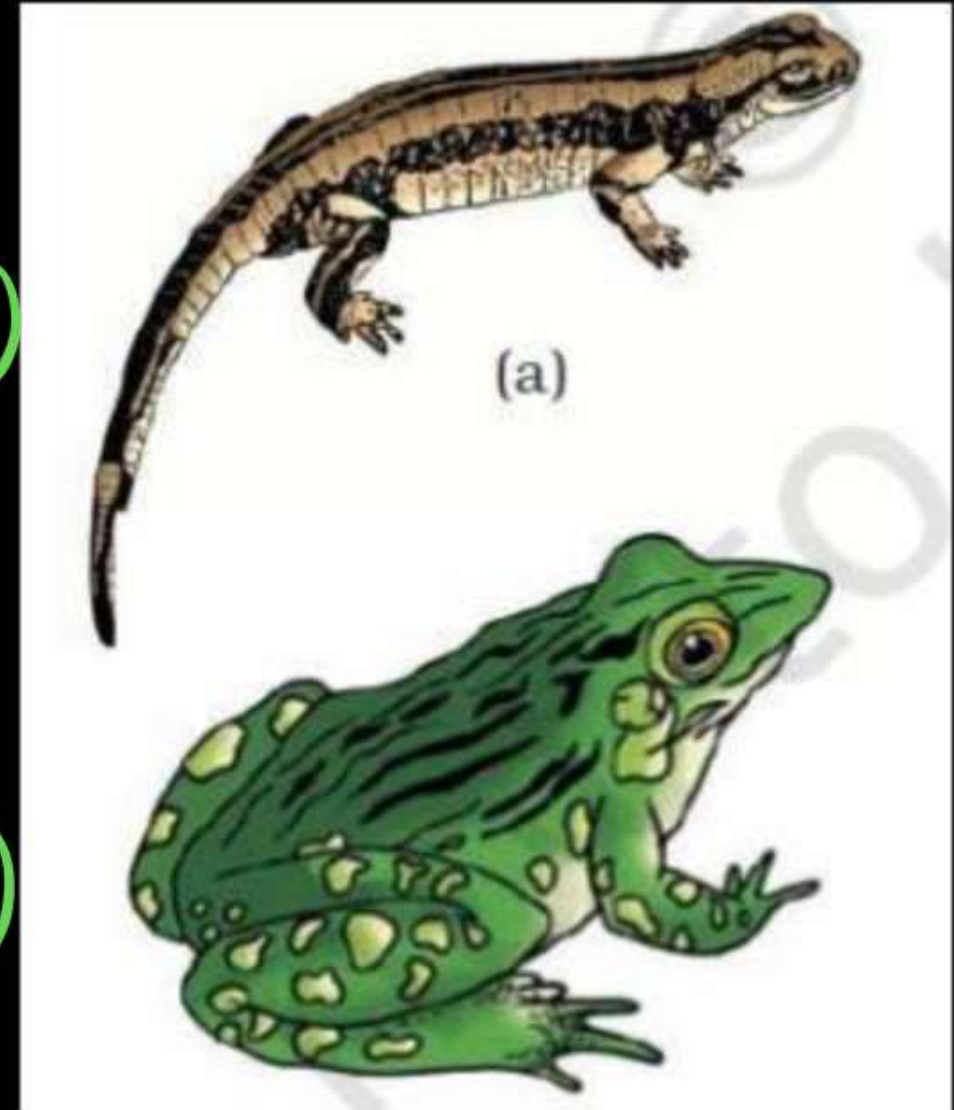


Figure 4.20 Examples of Bony fishes :
(a) Hippocampus (b) Catla

Amphibia :

1. Bufo.....^{Toad}.....,
2. Rana.....^{Frog}.....
3. Hyla.....^{Tree frog}.....
4. Salamandra.....^{Salamander}.....
5. Ichthyophis^{Limbless amphibian}.....
Q// =

(a)



Reptiles:

1. Chelone.....Turtle — (b)
2. Testudo.....Tortoise
3. Chameleon.....Tree lizard — (c)
4. Calotes.....Garden lizard
5. Crocodilus.....Crocodile — (a)
6. Alligator.....Alligator Hemidactylus.....Wall lizard
7. Poisonous snakes – Naja.....Cobra — (d)
Bangarus (Krait), Vipera (Viper).

(a)



(b)



(c)



(d)



Aves

1. Corvus.....Crow
2. Pigeon.....Columba
3. Psittacula.....Parrot (d)
4. Struthio.....Ostrich — (a)
5. Peacock.....Pavo (b)
6. Aptenodytes.....Penguin
7. Neophron.....Vulture (c)

(a)



(b)



(c)



(d)



Mammals

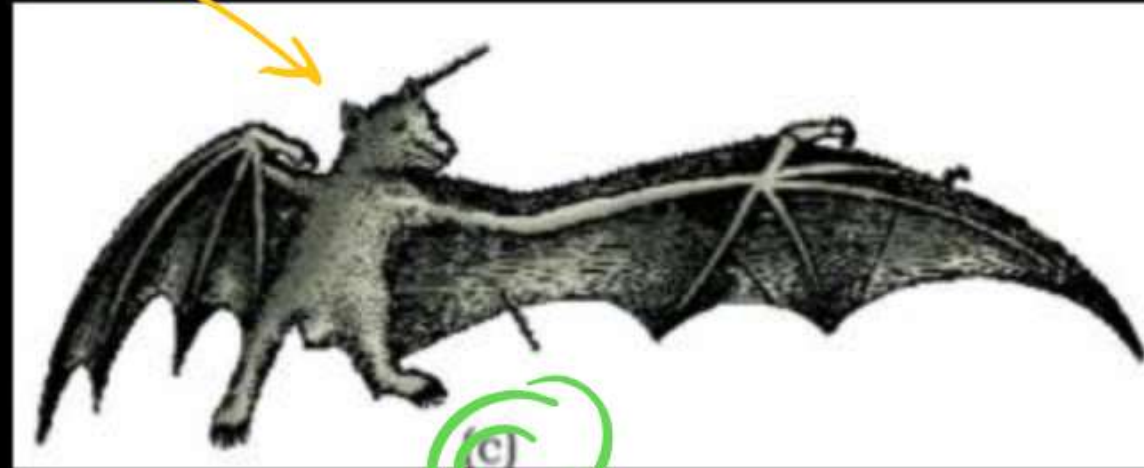
1. Ornithorhynchus..... *Platypus*
2. Macropus..... *Kangaroo*
3. Pteropus..... *Flying fox*
4. Balaenoptera..... *Whale*



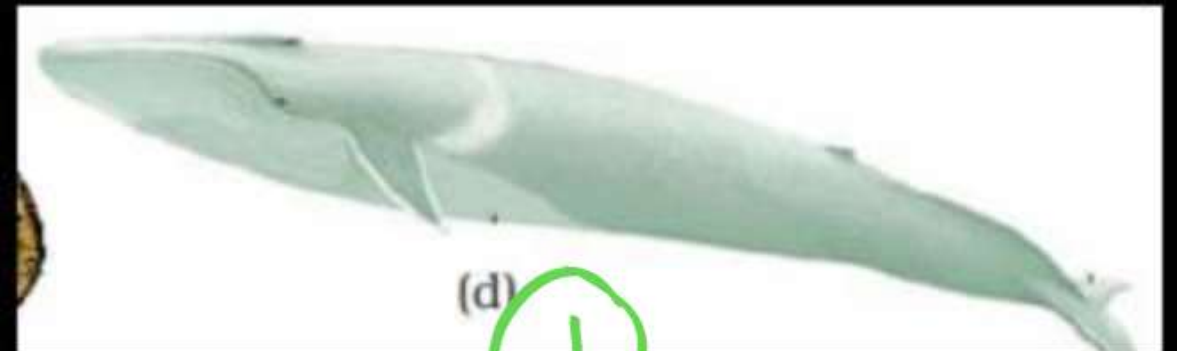
(a)



(b)



(c)



(d)

Frog

.

Body = Head + Trunk

Skin = Smooth & Slippery (Mucus + nt)

Cutaneous Respiration.

Nose - one pair

Ear = Internal Ear + Middle Ear + External Ear [Tympanum]

Dorsal = Olive Green

Ventral = Pale Yellow

4-Digits

Male Copulatory pad

Thorax + Abdomen

Trunk

Head

Eye

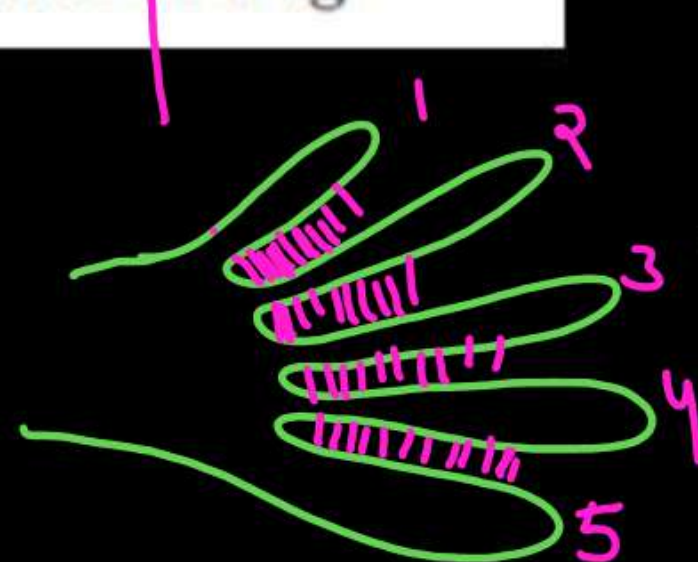
Fore limb

Hind limb

Figure 7.1 External features of frog

3rd Eye
↓
(Nictitating Membrane)

Neck & Tail } - nt



Frog

- **Rana tigrina** = The most common species of frog found in India.
- They do not have constant body temperature - **cold** blooded or poikilotherms. @
- Summer sleep = Aestivation. } H.W.
- Winter sleep = Hibernation.
- They have the ability to change the colour to hide them from their enemies = **camouflage**. This protective coloration is called = **mimicry**. @/ @/

Morphology of Frog

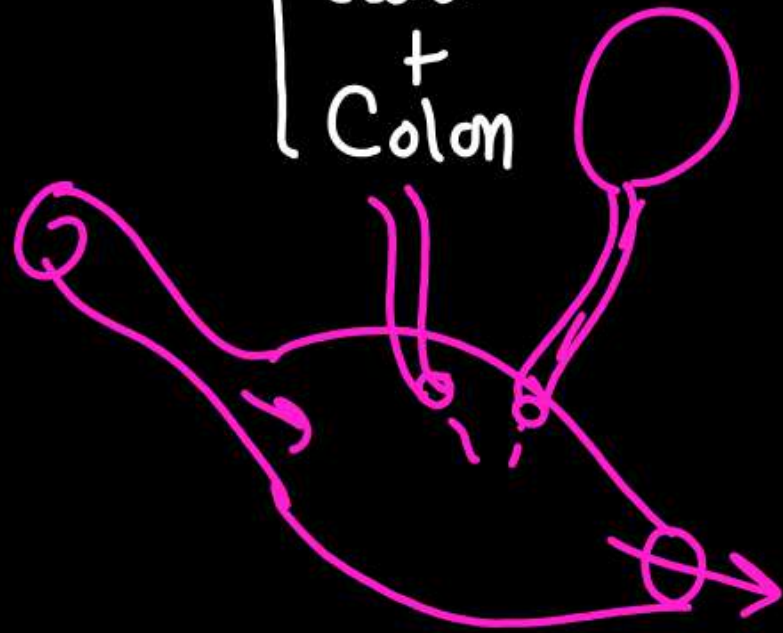
- The skin is smooth and slippery due to the presence of mucus. The skin is always maintained in a moist condition. = *Cutaneous Respiration*.
- Frog never drinks water but absorb it through the skin.
- Body of a frog is divisible into head and trunk =
- A neck and tail are absent
Above the mouth a pair of nostrils is present.
- Nictitating membrane
- Membranous tympanum (ear) =
- hind limbs end in = five digits ✓
- fore limbs that end in = four digits ✓
- Feet have webbed digits that help in swimming.
- Frogs exhibit sexual dimorphism, Male frogs - vocal sacs and copulatory pad on the first digit of the fore limbs.

Digestive System

1) Digestive Glands = Gastric, Pancreas, Liver

2) Digestive Tract = Mouth → Bilobed tongue 
[Carnivores = Small]

Intestine → Duodenum
- nt {
 - Jejunum
 - Ileum
 - Caecum
 - Colon



Stomach → HCl + Gastric juice } Chyme

Intestine → Duodenum
 ↓
 Ileum
 [Villi & Microvilli] = for absorption

Bile = Fat

Pancreatic juice = Protein + Carbohydrate

Rectum

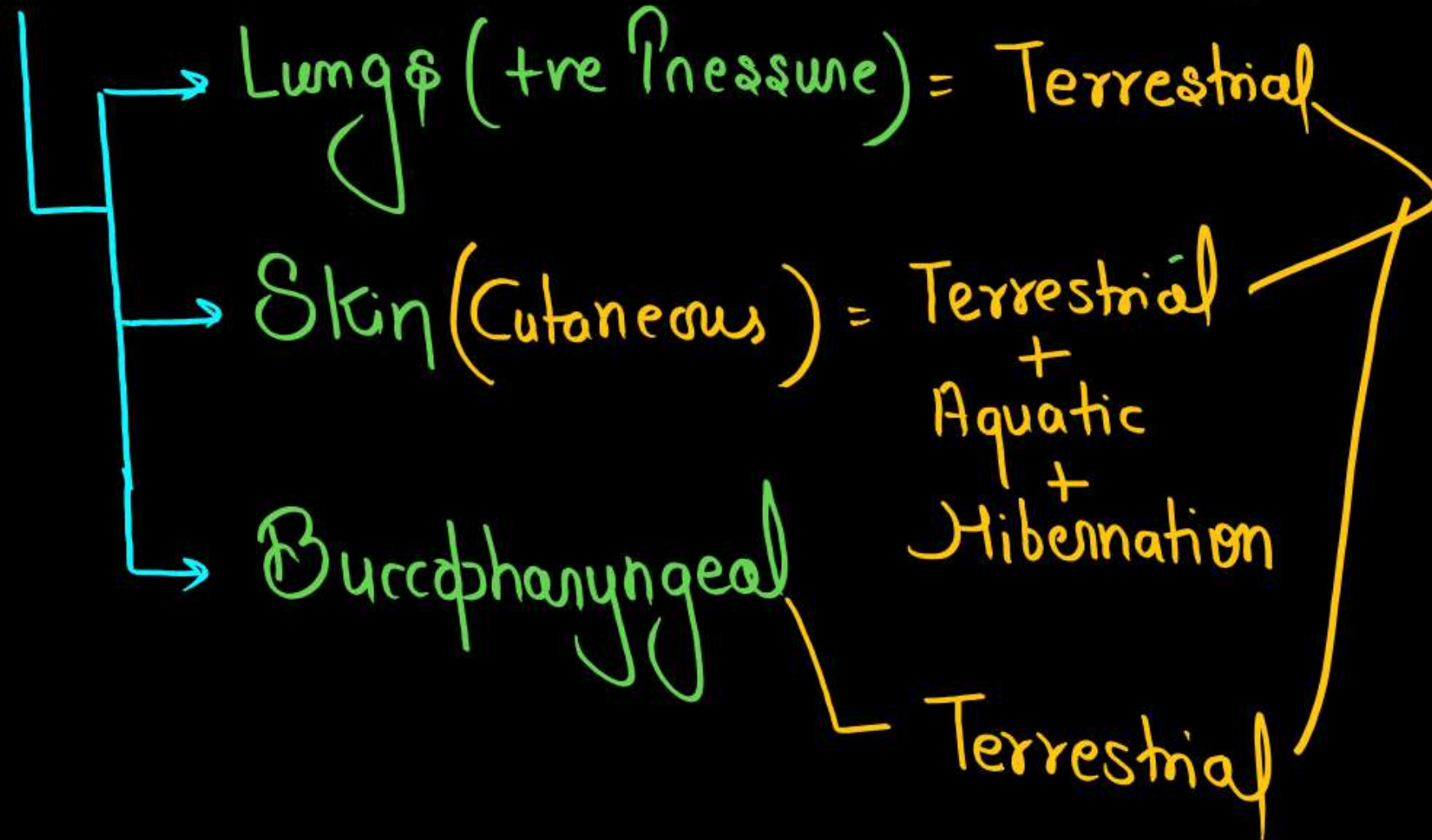
Cloaca [urine + sperm + undigested food]

Respiratory System

②

Respiratory Organs

(Adult)

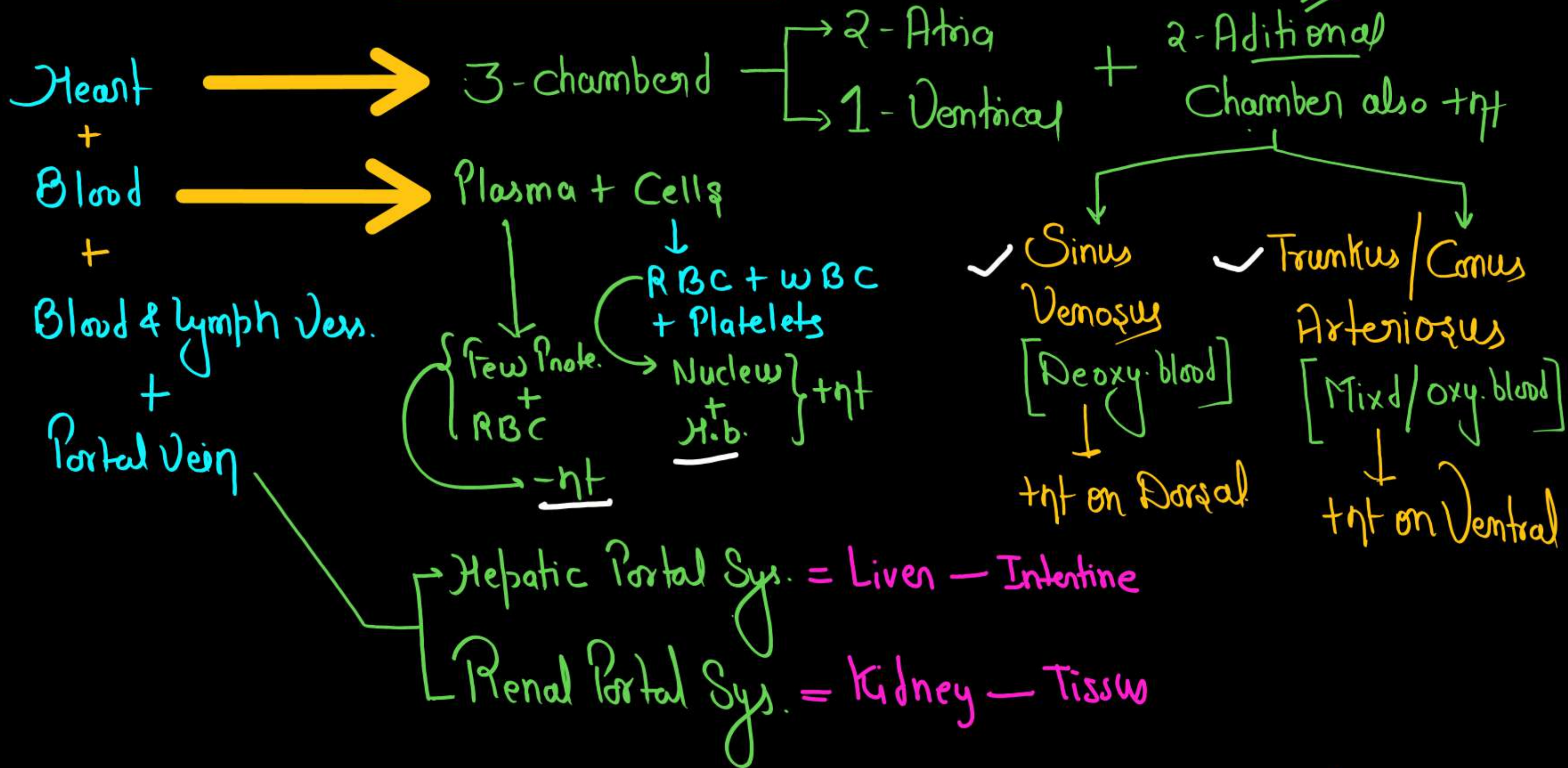


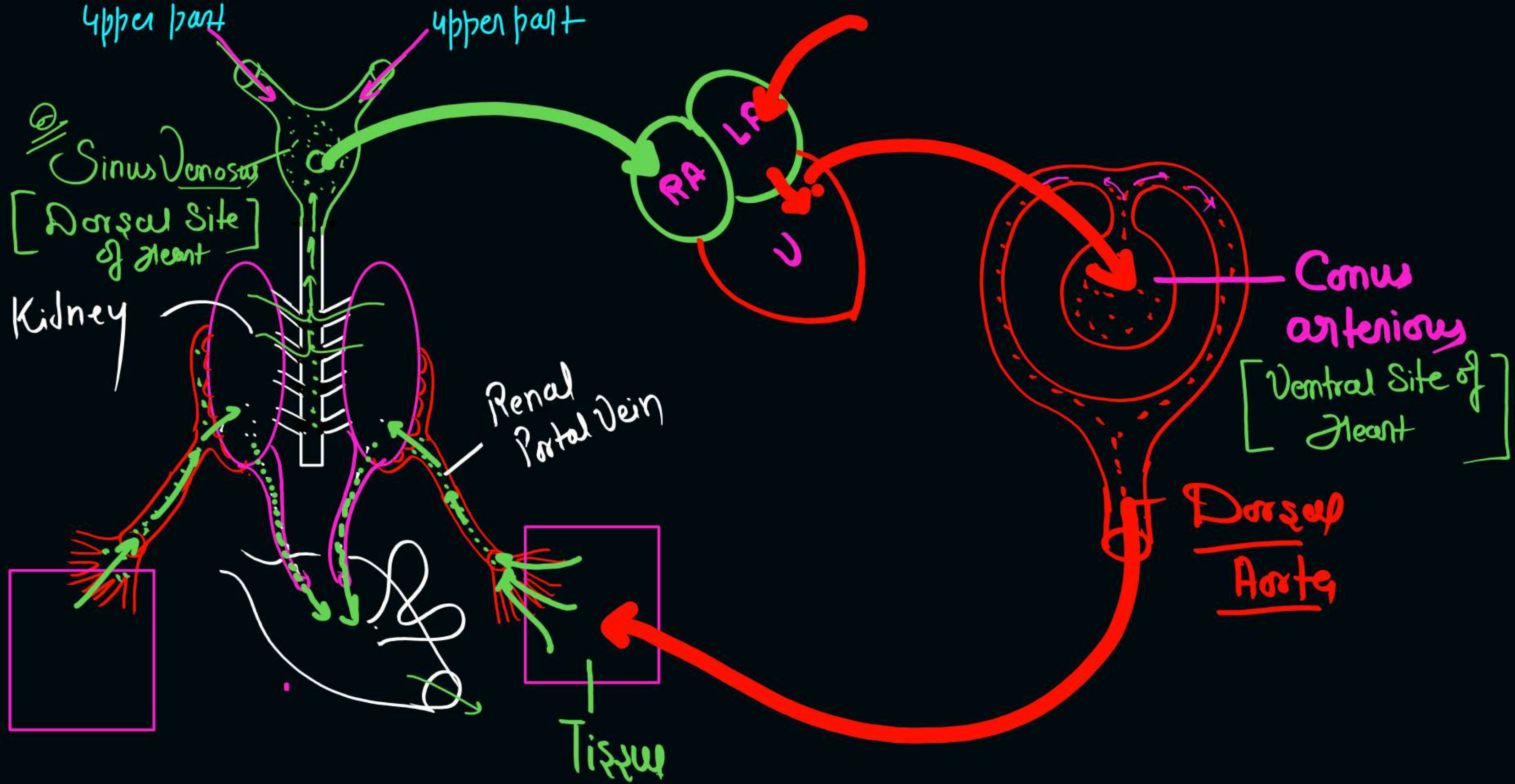
(Larva Tadpole)

Gills

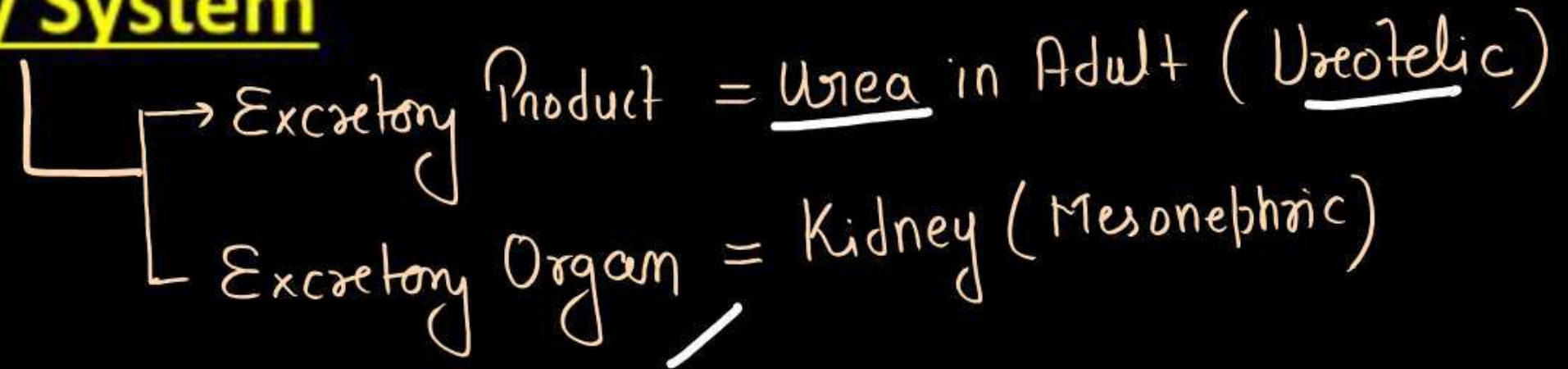
Circulatory System

Closed Type



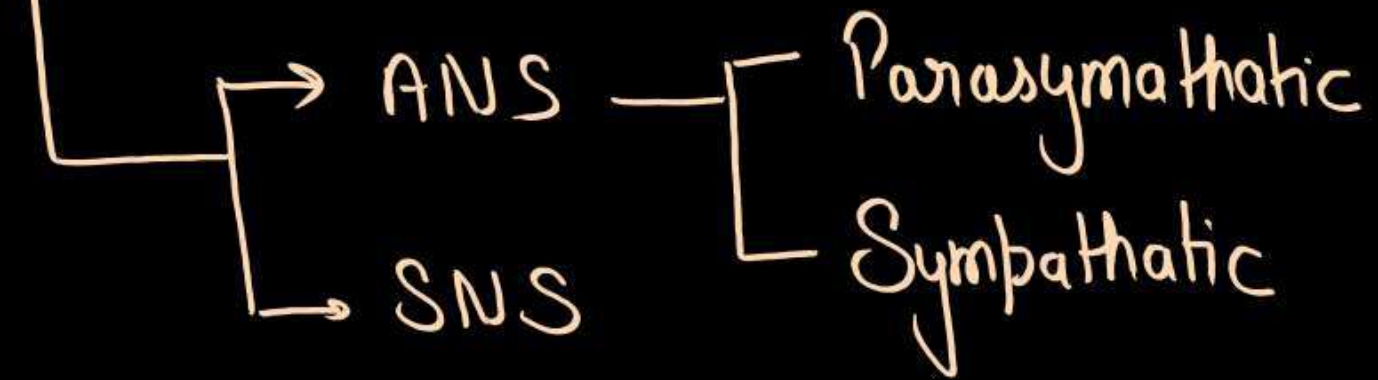


Excretory System



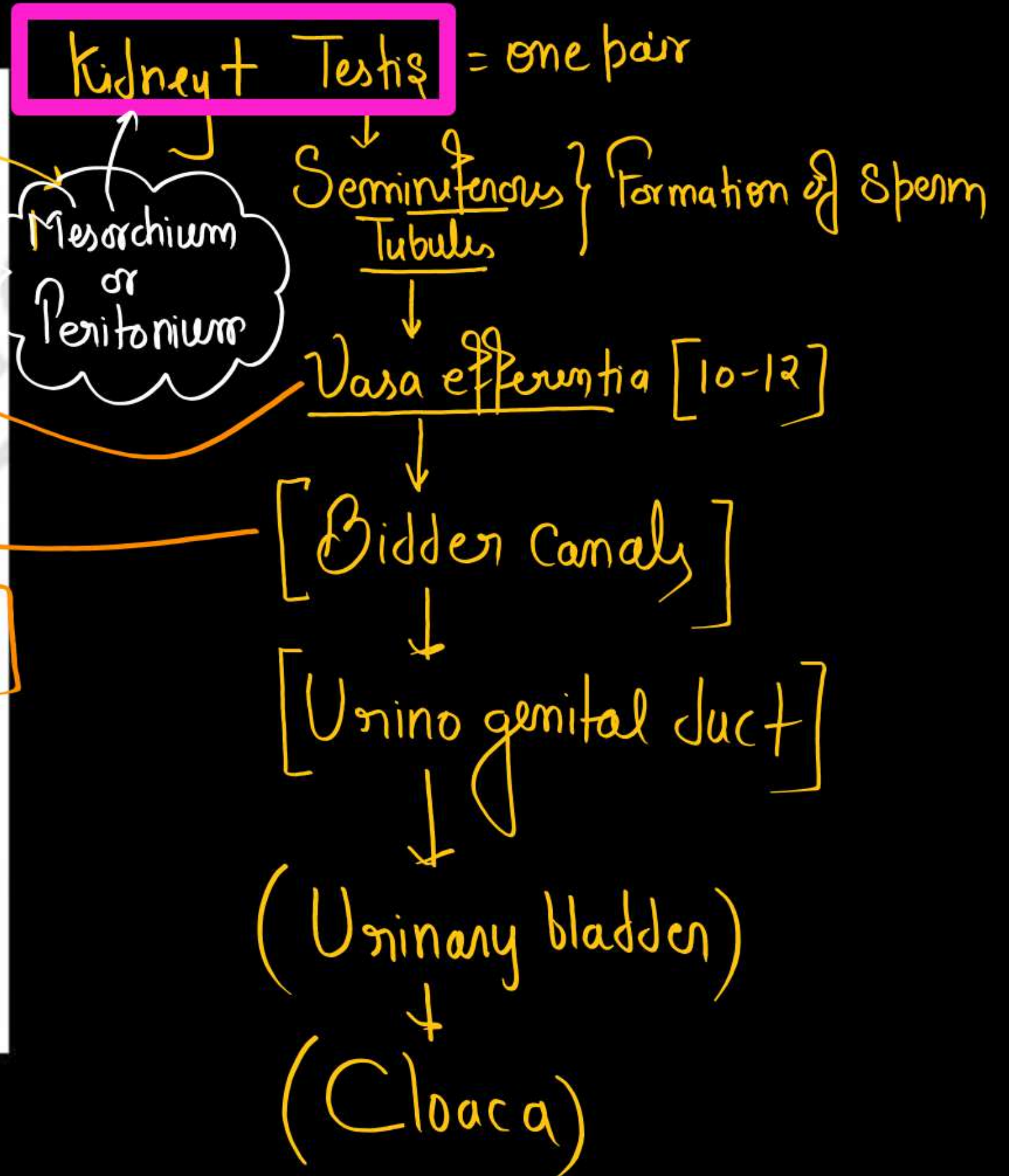
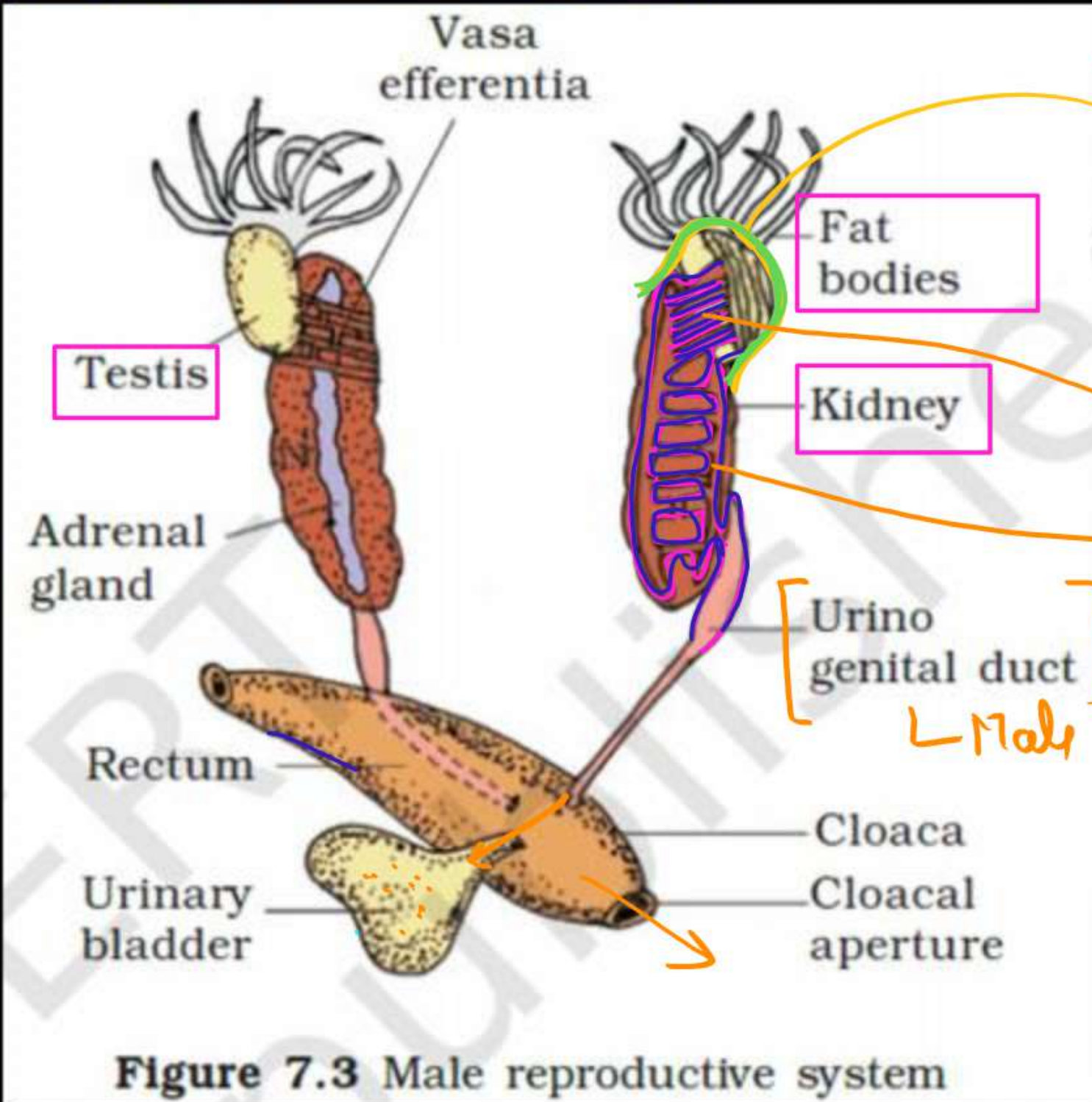
Nervous System →

- 1) - CNS = Brain + Spinal cord
- 2) - PNS = Cranial + Spinal Nerve
(10-Pairs) (10-Pairs)



Male Reproductive System

lvvmb



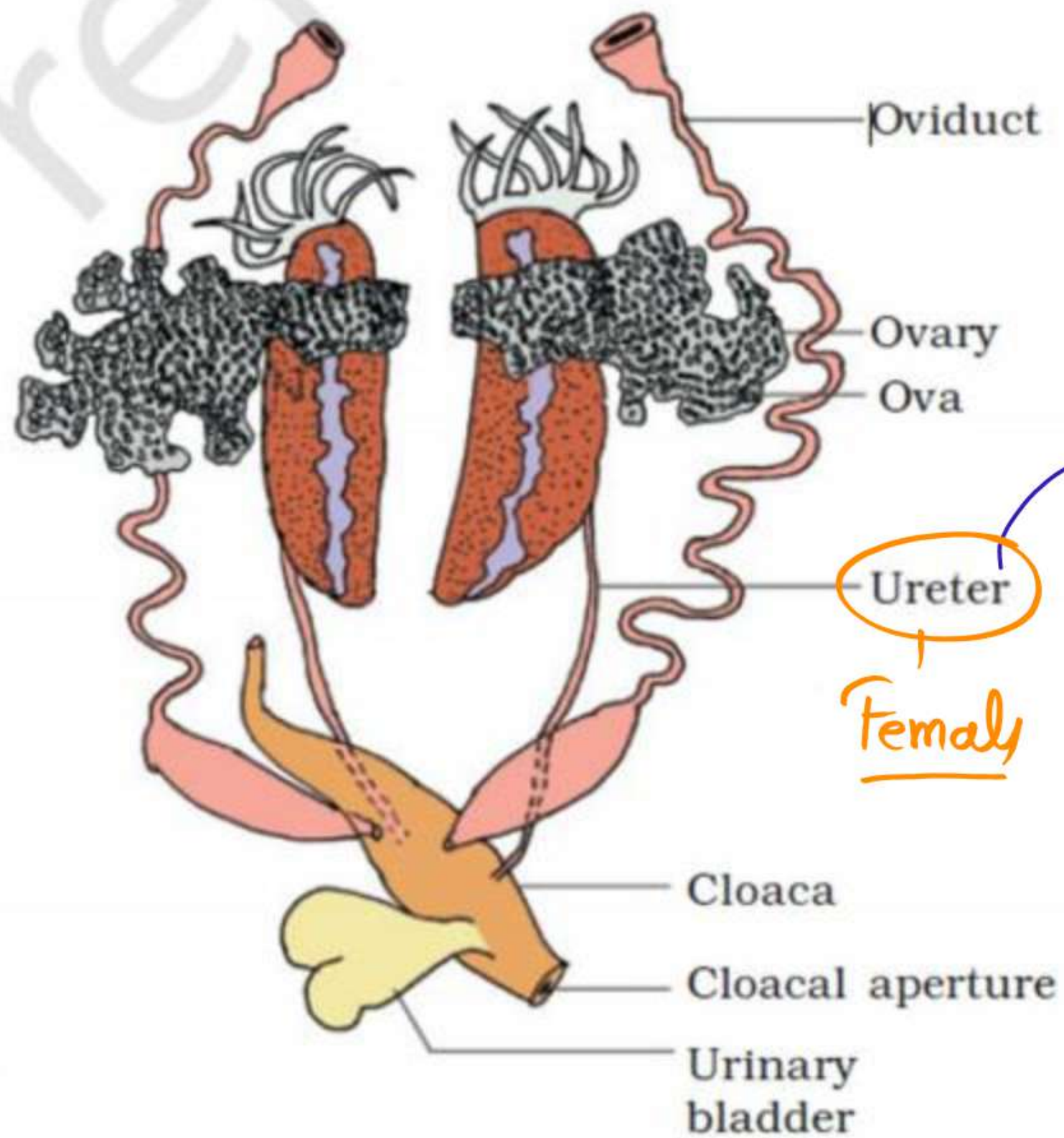
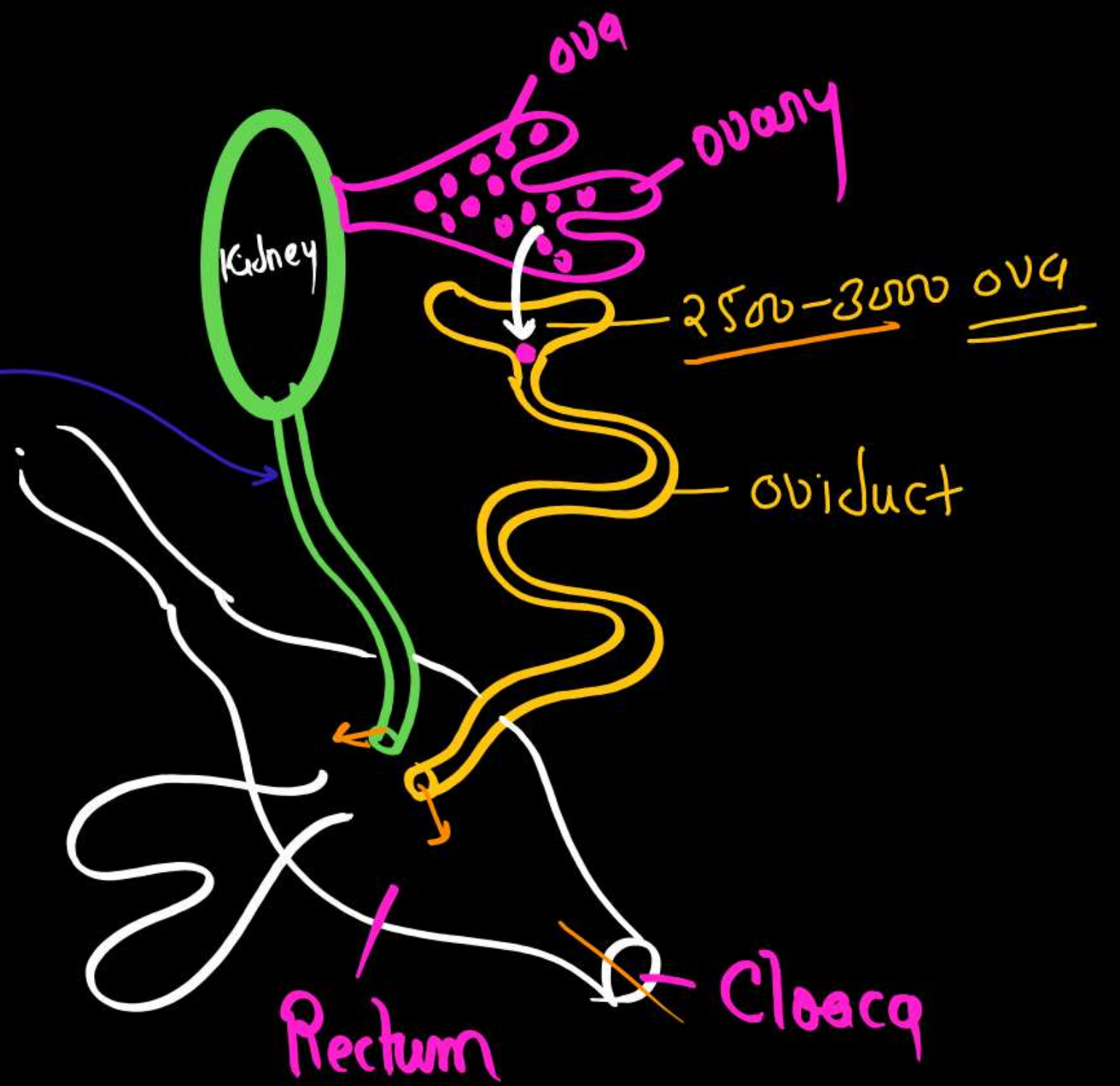


Figure 7.4 Female reproductive system



Read the following statements (1–3) and answer the question:

- (1) Tidal Volume (TV) is the amount of air inspired or expired during normal breathing, averaging about 500 mL.
- (2) A healthy adult breathes approximately 2,000–3,000 mL of air per minute during quiet breathing.
- (3) Residual Volume (RV) is the air left in the lungs after forced expiration, typically 1100–1200 mL.

How many of the above statements are correct?

- (1) Only 1 and 2
- (2) Only 2 and 3
- (3) Only 1 and 3
- (4) All correct

Under normal physical conditions in human beings every 100 ml oxygenated blood can deliver _____ ml of O_2 to the tissues.

- (a) 2 ml
- (c) 4 ml

- (b) 5 ml
- (d) 10 ml

(b)

(3)

Select the favorable conditions required for the formation of oxyhaemoglobin at the alveoli.

lungs

- (a) High pO_2 , low pCO_2 , less H^+ , lower temperature
- (b) Low pO_2 , high pCO_2 , more H^+ , higher temperature
- (c) High pO_2 , high pCO_2 , less H^+ , higher temperature
- (d) Low pO_2 , low pCO_2 , more H^+ , higher temperature

(a)

The partial pressures (in mmHg) of oxygen (O_2) and carbon dioxide (CO_2) at alveoli (the site of diffusion) are:

- (a) $pO_2=104$ and $pCO_2=40$
- (b) $pO_2=40$ and $pCO_2=45$
- (c) $pO_2=95$ and $pCO_2=40$
- (d) $pO_2=159$ and $pCO_2=0.3$

(a)

104
40

Match the items given in Column I with those in Column II and select the correct option given below:

Column I		Column II	
a.	Tidal volume	i.	2500-3000 mL
b.	Inspiratory Reserve volume	ii.	1100-1200 mL
c.	Expiratory Reserve volume	iii.	500-550 mL
d.	Residual volume	iv.	1000-1100 mL

(a) a-i, b-iv, c-ii, d-iii

(b) a-iii, b-i, c-iv, d-ii

(c) a-iii, b-ii, c-i, d-iv

(d) a-iv, b-iii, c-ii, d-i

Match List I with List II.

vmp

List I	List II
A. P Wave	I. Heart muscles are electrically silent.
B. QRS complex	II. Depolarisation of ventricles.
C. T wave	III. Depolarisation of atria.
D. T-P gap	IV. Repolarisation of ventricles.

(joint Dia.)

Choose the correct answer from the options given below.

- (a) A-III, B-II, C-IV, D-I
- (b) A-II, B-III, C-I, D-IV
- (c) A-IV, B-II, C-I, D-III
- (d) A-I, B-III, C-IV, D-II

Match List I with List II:

	List I		List II
A.	P-wave	I.	Beginning of systole
B.	Q-wave	II.	Repolarisation of ventricles
C.	QRS	III.	Depolarisation of complex atria
D.	T-wave	IV.	Depolarisation of ventricles

Choose the correct answer from the options given below:

- (a) A-IV, B-III, C-II, D-I
- (b) A-II, B-IV, C-I, D-III
- (c) A-I, B-II, C-III, D-IV
- (d) A-III, B-I, C-IV, D-II

Which enzyme is responsible for conversion of inactive fibrinogen to fibrin?

- (a) Epinephrine (b) Thrombokinase
(c) Thrombin (d) Renin



Match the following columns and select the correct option.

	Column-I		Column-II
(1)	Eosinophils	(i)	Immune response
(2)	Basophils	(ii)	Phagocytosis
(3)	Neutrophils	(iii)	Release histamine, destructive enzymes
(4)	Lymphocytes	(iv)	Release granules containing histamine

- (a) (1)-(iv); (2)-(i); (3)-(ii); (4)-(iii)
(b) (1)-(i); (2)-(ii); (3)-(iv); (4)-(iii)
(c) (1)-(ii); (2)-(i); (3)-(iii); (4)-(iv)
(d) (1)-(iii); (2)-(iv); (3)-(ii); (4)-(i)

The QRS complex in a standard ECG represents:

- (a) Depolarization of ventricles
- (b) Repolarization of ventricles
- (c) Repolarization of auricles
- (d) Depolarization of auricles

a

Chronic disorder in which alveolar walls are damaged due to which respiratory surface is decreased.

- (1) Pneumonia
- (2) Tuberculosis
- (3) Emphysema
- (4) Asthma

3

All the components of the nodal tissue are autoexcitable. Why does the SA node acts as normal pacemaker?

- (a) SA node has the lowest rate of depolarization
- (b) SA node is the only component to generate Threshold
- (c) Only SA node convey the action potential to others
- (d) SA node has the highest rate of depolarization

Find the correct descending order of percentage proportion of leucocytes in human blood.

- (1) Neutrophils $\rightarrow$ Basophils $\rightarrow$ Lymphocytes $\rightarrow$ Acidophils (Eosinophils) $\rightarrow$ Monocytes
- (2) Monocytes $\rightarrow$ Neutrophils $\rightarrow$ Lymphocytes $\rightarrow$ Acidophils $\rightarrow$ Basophils
- (3) Neutrophils $\rightarrow$ Lymphocytes $\rightarrow$ Monocytes $\rightarrow$ Acidophils $\rightarrow$ Basophils
- (4) Lymphocytes $\rightarrow$ Acidophils $\rightarrow$ Basophils $\rightarrow$ Neutrophils $\rightarrow$ Monocytes

Given below are statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Nephrons are of two types: cortical and juxtamedullary, based on their relative position in cortex and medulla.

Reason R: Juxtamedullary nephrons have short loop of Henle, whereas cortical nephrons have longer loop of Henle.

In the light of the above statements, choose the correct answer from the options given below:

- (a) Both A and R are true, but R is NOT the correct explanation of A.
- (b) A is true, but R is false.
- (c) A is false, but R is true.
- (d) Both A and R are true, and R is the correct explanation of A.

Erythropoietin hormone which stimulates R.B.C. formation is produced by

- (a) Alpha cells of pancreas
- (b) The cells of rostral adenohypophysis
- (c) The cells of bone marrow
- (d) Juxtaglomerular cells of the kidney

Which of the following statements are correct?

- A. An excessive loss of body fluid from the body switches off osmoreceptors. ~~X~~
- B. ADH facilitates water reabsorption to prevent diuresis. ~~/~~
- C. ANF causes vasodilation. ~~/~~
- D. ADH causes an increase in blood pressure. ~~/~~
- E. ADH is responsible for a decrease in GFR. ~~X~~

Choose the correct answer from the options given below:

- (a) ~~/~~ B, C and D only (b) A, B and E only
- (c) C, D and E only (d) A and B only

(a)

Which of the following would help in prevention of diuresis?

- (a) Atrial natriuretic factor causes vasoconstriction
- (b) Decrease in secretion of renin by JG cells
- (c) More water reabsorption due to under secretion of ADH C & D
- (d) Reabsorption of Na^+ and water from renal tubules due to aldosterone //

Presence of which of the following conditions in urine are indicative of Diabetes Mellitus?

- (a) Uremia and Ketonuria
- (b) Uremia and Renal Calculi
- (c) Ketonuria and Glycosuria C
- (d) Renal calculi and Hyperglycemia

13. Match the following parts of a nephron with their function:

Column I		Column II	
A	Descending limb of Henle's loop	i.	Reabsorption of salts only
B	Proximal Convoluted tubule	ii.	Reabsorption of water only
C	Ascending limb of Henle's loop	iii.	Conditional reabsorption of sodium ion and water
D	Distal convoluted tubule	iv.	Reabsorption of ion, water and organic nutrients

- (a) A-ii, B-iii, C-i, D-iv (b) A-i, B-ii, C-iii, D-iv
(c) A-ii, B-iv, C-i, D-iii (d) A-iv, B-i, C-ii, D-iii

15. Match the items given in column I with those in Column II and select the correct option given below:

Column I		Column II	
a.	Glycosuria	i.	Accumulation of uric acid in joints
b.	Gout	ii.	Mass of crystallized salts within the kidney
c.	Renal calculi	iii.	Inflammation in glomeruli
d.	Glomerular nephritis	iv.	Presence of glucose in urine

(a) a-ii, b-iii, c-i, d-iv

(b) a-i, b-ii, c-iii, d-iv

(c) a-iii, b-ii, c-iv, d-i

(d) a-iv, b-i, c-ii, d-iii

Which of the following statements is correct?

- (a) The ascending limb of loop of Henle is impermeable to water
- (b) The descending limb of loop of Henle is impermeable to water
- (c) The ascending limb of loop of Henle is permeable to water
- (d) The descending limb of loop of Henle is permeable to electrolytes

Match List I with List II:

List-I	List-II
A. Fibrous joints	I. Adjacent vertebrae, limited movement
B. Cartilaginous joints	II. Humerus and Pectoral girdle, rotational movement
C. Hinge	III. Skull, don't allow any movement
D. Ball and socket joints	IV. Knee, help in locomotion

Choose the correct answer from the options given below:

- (a) A-I, B-III, C-II, D-IV
- (b) A-II, B-III, C-I, D-IV
- (c) A-III, B-I, C-IV, D-II
- (d) A-IV, B-II, C-III, D-I

Match list 1 and list 2

Choose the correct answers from the options given below

	List - I		List - II
1.	Scapula	i.	Cartilaginous joints
2.	Cranium	ii.	Flat bone
3.	Sternum	iii.	Fibrous joints
4.	Vertebral column	iv.	Triangular flat bone

- (a) 1 – iv, 2 – ii, 3 – iii, 4 – i
(b) 1 – iv, 2 – iii, 3 – ii, 4 – i
(c) 1 – i, 2 – iii, 3 – ii, 4 – iv
(d) 1 – ii, 2 – iii, 3 – iv, 4 – i

During muscular contraction, which of the following events occur.

1. 'H' zone disappears
2. 'A' band widens
3. 'I' band reduces in width
4. Myosin hydrolysis ATP releasing ADP and P
5. Z line attaches to actin and pull inwards

- (a) 2,3,4 and 5 only
(b) 2,4,5 and 1 only
(c) 1,3,4 and 5 only
(d) 1,2,3 and 4 only
(e)

1. Match List I with List II:

List I	List II
A. Pons	I. Provides additional space for Neurons, regulates posture and balance
B. Hypothalamus	II. Controls respiration and gastric secretions
C. Medulla	III. Connects different regions of the brain
D. Cerebellum	IV. Neuro secretory cells

Choose the correct answer from the options given below:

- (a) A-III, B-IV, C-II, D-I
- (b) A-I, B-III, C-II, D-IV
- (c) A-II, B-I, C-III, D-IV
- (d) A-II, B-III, C-I, D-IV

The parts of human brain that helps in regulation of sexual behaviour, expression of excitement, pleasure, rage, fear, etc., are

- (a) Corpora quadrigemina and hippocampus
- (b) Brain stem and epithalamus
- (c) Corpus callosum and thalamus
- (d) Limbic system and hypothalamus

Which of the following structures or regions is incorrectly paired with its functions?

- (a) Hypothalamus: production of releasing hormones and regulation of temperature, hunger and thirst
- (b) Limbic system: consists of fibre tracts that interconnect different regions of brain; controls movement
- (c) Medulla oblongata: controls respiration and cardiovascular reflexes
- (d) Corpus callosum: band of fibers connecting left and right cerebral hemispheres

Glucagon is a peptide hormone secreted from the alpha cells of the pancreatic islets of Langerhans
Match List I with List II.

List I	List II
A. Exophthalmic goiter	I. Excess secretion of cortisol, moon face & hyperglycemia
B. Acromegaly	II. Hypo-secretion of thyroid hormone and stunted growth.
C. Cushing's syndrome	III. Hyper secretion of thyroid hormone & protruding eye balls.
D. Cretinism	IV. Excessive secretion of growth hormone.

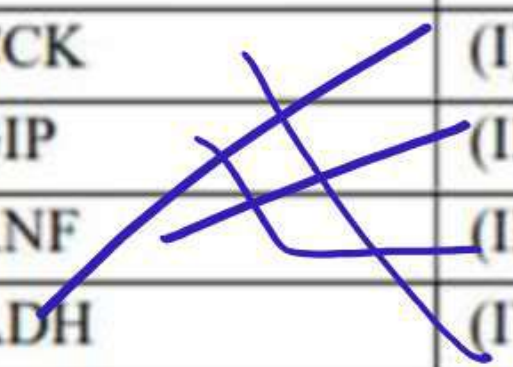
Choose the correct answer from the options given below:

- (a) A-IV, B-II, C-I, D-III
(b) A-III, B-IV, C-II, D-I
(c) A-III, B-IV, C-I, D-II
(d) A-I, B-III, C-II, D-IV

C

Match List I with List II.

List I	List II
(A) CCK	(I) Kidney
(B) GIP	(II) Heart
(C) ANF	(III) Gastric gland
(D) ADH	(IV) Pancreas



Choose the correct answer from the options given below:

- (a) A-III, B-II, C-IV, D-I
- (b) A-II, B-IV, C-I, D-III
- (c) A-IV, B-II, C-III, D-I
- (d) A-IV, B-III, C-II, D-I

Match the following columns and select the correct option.

Column I	Column II
(a) Pituitary gland	(i) Grave's disease
(b) Thyroid gland	(ii) Diabetes mellitus
(c) Adrenal gland	(iii) Diabetes insipidus
(d) Pancreas	(iv) Addison's disease

- (a) a – iii; b – i; c – iv; d – ii
(b) a – ii; b – i; c – iv; d – iii
(c) a – iv; b – iii; c – i; d – ii
(d) a – iii; b – ii; c – i; d – iv

Select the correct statement.

- (a) Insulin acts on pancreatic cells and adipocytes.
(b) Insulin is associated with hyperglycemia.
(c) Glucocorticoids stimulate gluconeogenesis.
(d) Glucagon is associated with hypoglycemia.

Match the column and select the correct option.

Column I	Column II
(a) Pituitary gland	(i) Steroid
(b) Epinephrine	(ii) Neuropeptide
(c) Endorphins	(iii) Peptide, protein
(d) Cortisol	(iv) Biogenic amines

(a) a-iv, b-i, c-ii, d-iv

(b) a-iii, b-iv, c-ii, d-i

(c) a-iv, b-iii, c-i, d-ii

(d) a-iii, b-iv, c-i, d-ii

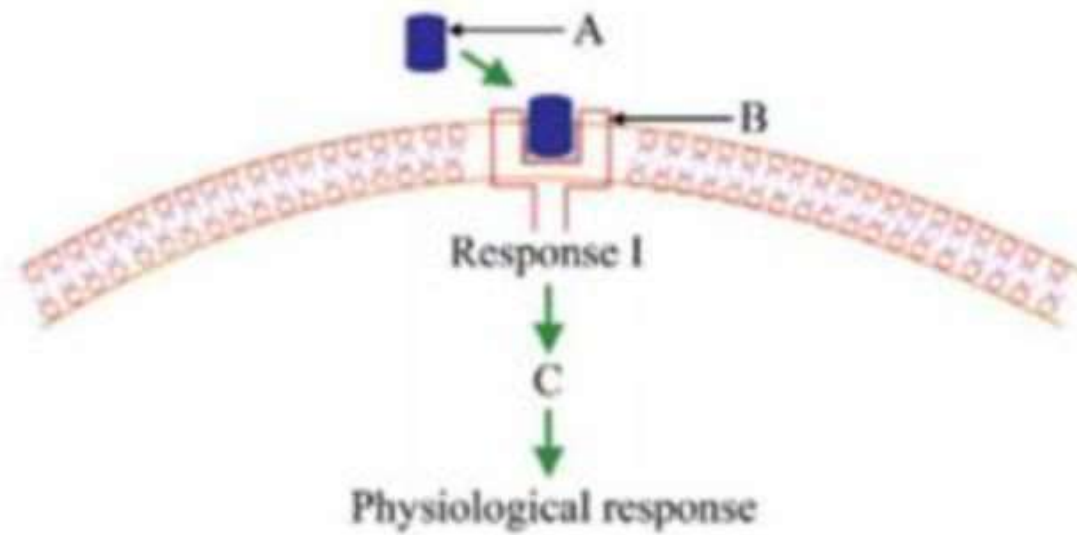
Match the following hormones with the respective disease.

- | | |
|---------------------|-------------------------|
| (p) Insulin | (i) Addison's disease |
| (q) Thyroxin | (ii) Diabetes insipidus |
| (r) Corticoids | (iii) Acromegaly |
| (s) Growth Hormones | (iv) Goitre |
| | (v) Diabetes mellitus |

Select the correct option.

- (a) $p \rightarrow (ii)$; $q \rightarrow (iv)$; $r \rightarrow (iii)$; $s \rightarrow (i)$
- (b) $p \rightarrow (v)$; $q \rightarrow (iv)$; $r \rightarrow (i)$; $s \rightarrow (iii)$
- (c) $p \rightarrow (ii)$; $q \rightarrow (iv)$; $r \rightarrow (i)$; $s \rightarrow (iii)$
- (d) $p \rightarrow (v)$; $q \rightarrow (i)$; $r \rightarrow (ii)$; $s \rightarrow (iii)$

Identify A, B and C in the diagrammatic representation of the mechanism of hormone action.



Select the correct option from the following.

- (a) A- steroid hormones, B- Hormone receptor complex, C- Protein
- (b) A- Protein hormone, B- Receptor, C- Cyclic AMP
- (c) A- Steroid hormone, B- Receptor, C- Second messenger
- (d) A- Protein hormone, B- Cyclic AMP, C- Hormone receptor complex

Which of the following statement is correct?

- (1) Essential oils like lemon grass oils are primary metabolites ~~X~~
- (2) Adenine and Guanine ~~X~~ are substituted pyrimidine
- (3) GLUT-4 enables glucose transport into cells
- (4) Proteins are linear chains of amino acids linked together by glycosidic ~~X~~ bonds

3

Match the following:

Column-I		Column-II	
1.	Primary structure	a.	Assembly of than one polypeptide.
2.	Secondary structure	b.	The sequence of amino acids.
3.	Tertiary structure	c.	Folded in the form of right handed helix.
4.	Quaternary structure	d.	Folded upon itself like a hollow woolen ball.

- (1) a-2, b-3, c-4, d-1
(2) a-1, b-4, c-3, d-2
(3) a-1, b-3, c-4, d-2
(4) a-2, b-4, c-1, d-3

Read the following statements (1 – 3) and answer the question which follows them:

1. Prosthetic groups are organic compounds and they are tightly bound to apoenzymes
2. In peroxidase and catalase, haem is the prosthetic group and it is a part of the active site of the enzyme
3. No enzymes require metal ions for their activity

How many of the above statements are correct?

- (1) Only 1 and 2
- (2) Only 2 and 3
- (3) Only 1, 2 and 3
- (4) All correct

1

Match List-I with List-II.

List-I

List-II

- | | |
|------------------|------------------|
| A. Adenosine | I. Nitrogen base |
| B. Adenylic acid | II. Nucleotide |
| C. Adenine | III. Nucleoside |
| D. Alanine | IV. Amino acid |

Choose the option with all correct matches.

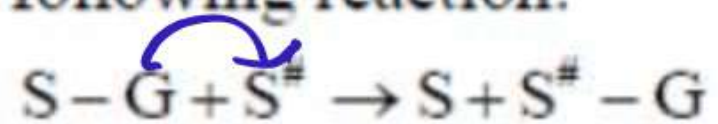
- (a) A-III, B-II, C-I, D-IV
- (b) A-II, B-III, C-I, D-IV
- (c) A-III, B-IV, C-II, D-I
- (d) A-III, B-II, C-IV, D-I

a

Which one of the following enzymes contains 'Haem' as the prosthetic group?

- (a) Succinate dehydrogenase
- ☒ (b) Catalase
- (c) RuBisCo
- (d) Carbonic anhydrase

Name the class of enzyme that usually catalyze the following reaction:



Where, $G \rightarrow$ a group other than hydrogen

$S \rightarrow$ a substrate

$S^{\#} \rightarrow$ another substrate

- | | |
|---|------------|
| ☒ (a) Transferase | (b) Ligase |
| (c) Hydrolase | (d) Lyase |

Inhibition of Succinic dehydrogenase enzyme by malonate is a classical example of

- (a) Feedback inhibition
- (b) Competitive inhibition
- (c) Enzyme activation
- (d) Cofactor inhibition

Choose the mismatch:

- (1) Dehydrogenase - catalyse oxidoreduction
- (2) Transferase - catalyses transfer of group
- (3) Hydrolase - enzyme catalyses hydrolysis of bonds like ester, ether etc.
- (4) Lyases - enzyme catalyses linking together of 2 compounds

Match List I with List II:

List I	List II
A. Lipase	I. Peptide bond
B. Nuclease	II. Ester bond
C. Protease	III. Glycosidic bond
D. Amylase	IV. Phosphodiester bond

Choose the correct answer from the options given below:

- (a) A-III, B-II, C-I, D-IV
(b) ~~A-II~~, ~~B-IV~~, C-I, D-III
(c) A-IV, B-I, C-III, D-II
(d) A-IV, B-II, C-III, D-I

(b)

Match the following columns:

Column I		Column II	
A.	Fatty acid	i.	Polymer of fructose
B.	Phospholipid	ii.	Modified sugar
C.	Glucosamine	iii.	Lecithin
D.	Inulin	iv.	Palmitic acid

- (1) A–(i), B–(ii), C–(iii), D–(iv)
(2) A–(iv), B–(iii), C–(ii), D–(i)
(3) A–(ii), B–(iii), C–(iv), D–(i)
(4) A–(iii), B–(iv), C–(i), D–(ii)

Malonate inhibits the growth of pathogenic bacteria by inhibiting the activity of

- (a) Amylase
(b) Lipase
(c) Dinitrogenase
(d) Succinic dehydrogenase

Match list I with list II. Choose the correct answer from the options given below:

	List I		List II
(a)	Glycogen	(i)	Hormone
(b)	Globulin	(ii)	Biocatalyst
(c)	Steroids	(iii)	Antibody
(d)	Thrombin	(iv)	Storage product

(a) (a) - (iii); (b) - (ii); (c) - (iv); (d) - (i)

(b) (a) - (iv); (b) - (ii); (c) - (i); (d) - (iii)

(c) (a) - (ii); (b) - (iv); (c) - (iii); (d) - (i)

(d) (a) - (iv); (b) - (iii); (c) - (i); (d) - (ii)

Identify the incorrect pair.

(a) Lectins- Concanavalin A

(b) Drugs- Ricin

(c) Alkaloids- Codeine

(d) Toxin- Abrin

6

Match list-I with list-II.

List-I		List-II	
(a)	Protein	(i)	C=C bonds
(b)	Unsaturated fatty acids	(ii)	Phosphodiester bonds
(c)	Nucleic acids	(iii)	Glycosidic bonds
(d)	Polysaccharides	(iv)	Peptide bonds

(a) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

(b) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

(c) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

(d) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)

Secondary metabolites such as nicotine, strychnine, and caffeine are produced by plants for their

- | | |
|---------------------|----------------------------|
| (a) Defense action | (b) Effect on reproduction |
| (c) Nutritive value | (d) Growth response |

Match the following.

- | | |
|-------------------------------------|---------------|
| (a) Inhibitor of catalytic activity | (i) Ricin |
| (b) Possess peptide bonds | (ii) Malonate |
| (c) Cell wall material in fungi | (iii) Chitin |
| (d) Secondary metabolite | (iv) Collagen |

Choose the correct option from the following.

- (a) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
(b) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)
(c) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)
(d) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

Role of the water vascular system in Echinoderms is:

- A. Respiration and Locomotion
- B. Excretion and Locomotion
- C. Capture and transport of food
- D. Digestion and Respiration
- E. Digestion and Excretion

Choose the correct answer from the options given below:

- | | |
|------------------|---------------------|
| (a) B and C Only | (b) B, D and E Only |
| (c) A and B Only | (d) A and C Only |

All living members of the class Cyclostomata are:

- | | |
|-----------------|------------------|
| (a) Symbiotic | (b) Ectoparasite |
| (c) Free living | (d) Endoparasite |

The unique mammalian characteristics are

- (a) Hairs, pinna and mammary glands
- (b) Hairs, pinna and indirect development
- (c) Pinna, monocondylic skull and mammary glands
- (d) Hairs, tympanic membrane and mammary glands

Match List I with List II:

List I	List II
A. Pleurobrachia	I. Mollusca
B. Radula	II. Ctenophora
C. Stomochord	III. Osteichthyes
D. Air bladder	IV. Hemichordata

Choose the correct answer from the options given below:

- (a) A-II, B-I, C-IV, D-III
- (b) A-II, B-IV, C-I, D-III
- (c) A-IV, B-III, C-II, D-I
- (d) A-IV, B-II, C-III, D-I

Match List I with List II:

List I	List II
A. <i>Pterophyllum</i>	I. Hag fish
B. <i>Myxine</i>	II. Saw fish
C. <i>Pristis</i>	III. Angel fish
D. <i>Exocoetus</i>	IV. Flying fish

Choose the correct answer from the options given below:

- (a) A-III, B-I, C-II, D-IV
- (b) A-IV, B-I, C-II, D-III
- (c) A-III, B-II, C-I, D-IV
- (d) A-II, B-I, C-III, D-IV

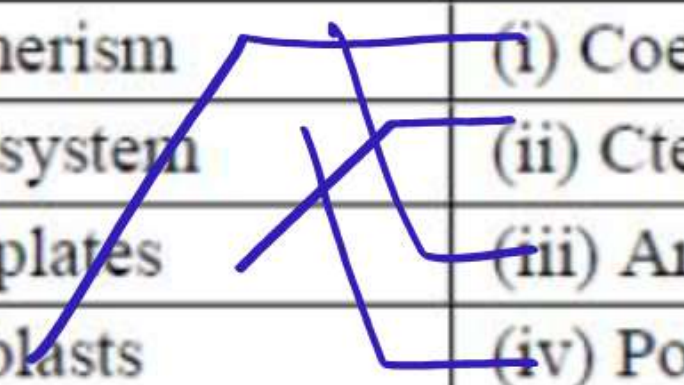
Match the following:

Column I	Column II
(p) <i>Physalia</i>	(i) Pearl oyster
(q) <i>Limulus</i>	(ii) Portuguese man of war
(r) <i>Ancylostoma</i>	(iii) Living fossil
(s) <i>Pinctada</i>	(iv) Hookworm

- (a) $p \rightarrow \text{(iii)}$; $q \rightarrow \text{(iv)}$; $r \rightarrow \text{(ii)}$; $s \rightarrow \text{(i)}$
(b) $p \rightarrow \text{(iii)}$; $q \rightarrow \text{(ii)}$; $r \rightarrow \text{(iv)}$; $s \rightarrow \text{(i)}$
(c) $p \rightarrow \text{(iii)}$; $q \rightarrow \text{(ii)}$; $r \rightarrow \text{(i)}$; $s \rightarrow \text{(iv)}$
(d) $p \rightarrow \text{(ii)}$; $q \rightarrow \text{(iii)}$; $r \rightarrow \text{(iv)}$; $s \rightarrow \text{(i)}$

Match the following columns and select the correct option.

Column I	Column II
(p) Metamerism	(i) Coelenterata
(q) Canal system	(ii) Ctenophora
(r) Comb plates	(iii) Annelida
(s) Cnidoblasts	(iv) Porifera



- (a) $p \rightarrow (i)$; $q \rightarrow (ii)$; $r \rightarrow (iii)$; $s \rightarrow (iv)$
(b) $p \rightarrow (iii)$; $q \rightarrow (iv)$; $r \rightarrow (i)$; $s \rightarrow (ii)$
(c) $p \rightarrow (iii)$; $q \rightarrow (iv)$; $r \rightarrow (ii)$; $s \rightarrow (i)$
(d) $p \rightarrow (iv)$; $q \rightarrow (iii)$; $r \rightarrow (i)$; $s \rightarrow (ii)$

Match the following columns and select the correct option.

Column I	Column II
(a) 6-15 pairs of gill slits	(i) <i>Trygon</i>
(b) Heterocercal caudal fin	(ii) Cyclostomes
(c) Air bladder	(iii) Chondrichthyes
(d) Poison sting	(iv) Osteichthyes

- (a) a – iv; b – ii; c – iii; d – i
 (b) a – i; b – iv; c – iii; d – ii
 (c) a – ii; b – iii; c – iv; d – i
 (d) a – iii; b – iv; c – i; d – ii

Match the following columns and select the correct option.

Column I	Column II
(a) Gregarious polyphagous pest	(i) <i>Asterias</i>
(b) Adult with radial symmetry and larva with bilateral symmetry	(ii) Scorpion
(c) Book lungs	(iii) <i>Ctenoplana</i>
(d) Bioluminescence	(iv) <i>Locusta</i>

- (a) a – iii; b – ii; c – i; d – iv
 (b) a – ii; b – i; c – iii; d – iv
 (c) a – i; b – iii; c – ii; d – iv
 (d) a – iv; b – i; c – ii; d – iii

Which of the following statements are true for the Phylum Chordata?

- (i) In Urochordata, notochord extends from head to tail, and it is present throughout their life.
- (ii) In Vertebrata, notochord is present during the embryonic period only.
- (iii) Central nervous system is dorsal and hollow.
- (iv) Chordata is divided into 3 subphyla:
Hemichordata, Tunicata and Cephalochordata.

- | | |
|--------------------|--------------------|
| (a) (i) and (ii) | (b) (ii) and (iii) |
| (c) (iv) and (iii) | (d) (iii) and (i) |

Match the following organisms with their respective characteristics.

- | | |
|--------------------------|-------------------------|
| (p) <i>Pila</i> | (i) Flame cells |
| (q) <i>Bombyx</i> | (ii) Comb plates |
| (r) <i>Pleurobrachia</i> | (iii) Radula |
| (s) <i>Taenia</i> | (iv) Malpighian tubules |

- (a) $p \rightarrow \text{(iii)}$; $q \rightarrow \text{(iv)}$; $r \rightarrow \text{(ii)}$; $s \rightarrow \text{(i)}$
- (b) $p \rightarrow \text{(ii)}$; $q \rightarrow \text{(iv)}$; $r \rightarrow \text{(iii)}$; $s \rightarrow \text{(i)}$
- (c) $p \rightarrow \text{(iii)}$; $q \rightarrow \text{(ii)}$; $r \rightarrow \text{(iv)}$; $s \rightarrow \text{(i)}$
- (d) $p \rightarrow \text{(iii)}$; $q \rightarrow \text{(ii)}$; $r \rightarrow \text{(i)}$; $s \rightarrow \text{(iv)}$

